



Agent-in-the-loop to distill expert knowledge into artificial intelligence models: a survey

Jiayuan Gao^{1,2} · Yingwei Zhang^{1,2} · Yiqiang Chen^{1,2} · Yihan Dong³ · Yuanzhe Chen^{1,2} · Shuchao Song^{1,2} · Boshi Tang⁴ · Yang Gu^{1,2}

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Abstract

Large-scale neural networks have revolutionized many general knowledge areas (e.g., computer vision and language processing), but are still rarely applied in many expert knowledge areas (e.g., healthcare), due to data sparsity and high annotation expenses. Human-in-the-loop machine learning (HIL-ML) incorporates expert domain knowledge into the modeling process, effectively addressing these challenges. Recently, some researchers have started using large models to substitute for certain tasks typically performed by humans. Although large models have limitations in expert knowledge areas, after being trained on trillions of examples, they have demonstrated advanced capabilities in reasoning, semantic understanding, grounding, and planning. These capabilities can serve as proxies of human, which introduces new opportunities and challenges in HIL-ML area. Based on the above, we summarize a more comprehensive framework, Agent-in-the-Loop Machine Learning (AIL-ML), where agent represents both humans and large models. AIL-ML can efficiently collaborate human and large model to construct vertical AI models with lower costs. This paper presents the first review of recent advancements in this area. First, we provide a formal definition of AIL-ML and discuss its related fields. Then, we categorize the AIL-ML methods based on data processing and model development, providing formal definitions for each, and present representative works in detail for each category. Third, we highlight relative applications of AIL-ML. Finally, we summarize the current literature and highlight future research directions.

Keywords Human-in-the-Loop · Machine learning · Deep learning · Large language models

1 Introduction

Over the past decade, large-scale neural networks (LeCun et al. 2015) have significantly advanced the development of computer vision, natural language processing, and other general knowledge domains. These breakthroughs, such as applications in image recognition (Krizhevsky et al. 2012), have enabled technologies like autonomous vehicles and facial

recognition systems. In the realm of natural language processing, large models like the GPT (Radford et al. 2018) series and BERT (Kenton and Toutanova 2019) have achieved near-human capabilities in generating and understanding complex texts. These models have not only enhanced the accuracy of machine translation but also improved interactions between users and artificial systems. Furthermore, the adoption of these technologies has driven innovations in intelligent customer service, showcasing their commercial potential.

Despite the remarkable achievements of large-scale neural networks in general knowledge areas, they face numerous challenges in expert domains such as healthcare, law, and so on. These fields often involve highly specialized and sparse datasets that are expensive to acquire and require extensive preprocessing and precise manual annotation to ensure data quality and usability. Additionally, traditional deep learning models are severely tested in their generalization capabilities within these domains (Zhang et al. 2021). The inability of training data to fully represent the complexities of the real world often means that these models may fail to accurately predict unseen scenarios, thereby limiting their reliability in professional applications. Data privacy becomes crucial when dealing with sensitive personal details in medical or legal contexts (Dwork 2006), demanding extra measures to satisfy regulatory and ethical rules. These challenges demonstrate the need to integrate expert domain knowledge into AI systems, where high accuracy and explainable decision-making processes are necessary for gaining end-user trust. For instance, a medical diagnostic system must be able to clearly justify its recommendations to physicians, allowing them to make informed clinical decisions. Thus, integrating the knowledge and judgment of human experts into the AI model becomes essential.

Facing the challenges above, Human-in-the-Loop Machine Learning (HIL-ML) (Wu et al. 2022) offers an effective solution by incorporating human knowledge into the machine learning process. It also emphasizes iterative interaction between humans and systems. As shown in Fig. 1, in the HIL-ML framework, humans contribute at all stages of the machine learning loop. These contributions enable machines to integrate human knowledge and experience, thereby enhancing model accuracy and adaptability. HIL-ML distills human expertise into machine learning through a dynamic feedback mechanism. Unlike static pipelines, the framework employs an adaptive loop architecture where data and model influence each other iteratively. These human interventions allow machines to integrate human knowledge and experience, thereby enhancing model accuracy, adaptability, and performance on unseen data. By directly distilling deep expert knowledge into models, HIL-ML not only improves model performance but also enhances explainability, making the models more transparent and trustworthy for end-users. This approach also reduces potential privacy risks when handling sensitive information, as human experts can directly monitor and adjust the processing of such data. Therefore, HIL-ML not only addresses the limitations of

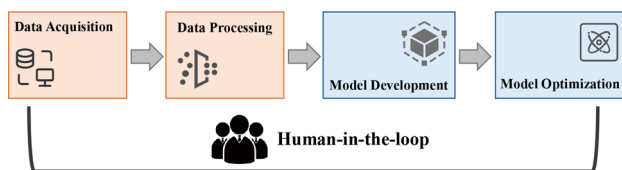


Fig. 1 The HIL-ML workflow which can be segmented into four main stages: Data Acquisition, Data Processing, Model Development, and Model Optimization, with continuous human involvement throughout the process to ensure iterative refinement and optimization of the models

traditional automation methods in professional fields but also enhances the reliability and effectiveness of AI systems in practical operations. By integrating human expert knowledge and feedback, HIL-ML enables AI systems to excel both in technological sophistication and practical application. Through human involvement, HIL-ML utilizes human knowledge for corrections and optimizations at various stages of modeling process, resulting in more precise and reliable machine learning models (Mosqueira-Rey et al. 2023a).

Recently, researchers have begun to employ large models to perform specific tasks traditionally executed by humans (Ding et al. 2022; Hsieh et al. 2023a). Although these large models hold some limitations in domains requiring expert knowledge, they have demonstrated advanced capabilities in reasoning, semantic understanding, grounding, and planning after being trained on trillions of examples. Their deep contextual understanding enables them to generate and comprehend language like humans (Mondorf and Plank 2024). Large models, such as GPT (Radford et al. 2018) and LLaMA (Touvron et al. 2023), possess deep contextual understanding and can generate meaningful outputs with minimal human input. These capabilities allow large models to act as proxies for human intelligence, thereby introducing new opportunities and challenges in the HIL-ML area. However, Current reviews (Wu et al. 2022; Mosqueira-Rey et al. 2023a; Xin et al. 2018) about HIL-ML fail to explore how to address these challenges. Moreover, the potential role of Large Models (Brown et al. 2020) as participants in HIL-ML has been largely overlooked. Building on these insights, we propose a novel concept called Large-Model-in-the-Loop Machine Learning (LMIL-ML). In the LMIL-ML framework, large models are integrated into the modeling loop, intervening at stages like data preprocessing or model development. By embedding large models into the training process, LMIL-ML reduces the reliance on human annotation. This framework achieves cost-efficient model training and improved accuracy by distilling large model expertise into more task-specific machine learning models.

Based on the frameworks of HIL-ML and LMIL-ML, we further introduce a unified and comprehensive framework, Agent in the Loop Machine Learning (AIL-ML). In this framework, agents include both humans and large models. This framework aims to leverage the complementary strengths of human cognitive skills and machine efficiency. AIL-ML combines human intuition and expertise with the computational power and reasoning ability of large models, creating a balance that leverages the strengths of both. These agents interact with the model at various stages-data processing, model training, and optimization-forming a machine learning loop. This dynamic process enhances the model's adaptability to changing environments, improves predictive accuracy across diverse tasks, and reduces the cost of iterative development. This paper provides a comprehensive overview of recent advancements in this area.

This study employs an extensive literature survey method to ensure comprehensive and high-quality research. We have reviewed works published at top computer science and human-computer interaction conferences from 2018 to 2024, such as IJCAI, ACL, KDD, AAAI, NIPS, ICML, CVPR, ICLR, CHI, IMWUT, CSCW and so on. Additionally, we have also investigated the latest research on arXiv from 2023 and 2024. By reading titles and abstracts, we selected the most relevant papers, classified them, and conducted in-depth analyses to ensure a precise understanding of the methods. As shown in Fig. 2, the paper begins by defining the core processes of AIL-ML and provides a structured framework for understanding its mechanisms. We then discuss the methodologies used in AIL-ML, emphasizing the critical aspects of data acquisition and processing as well as model development

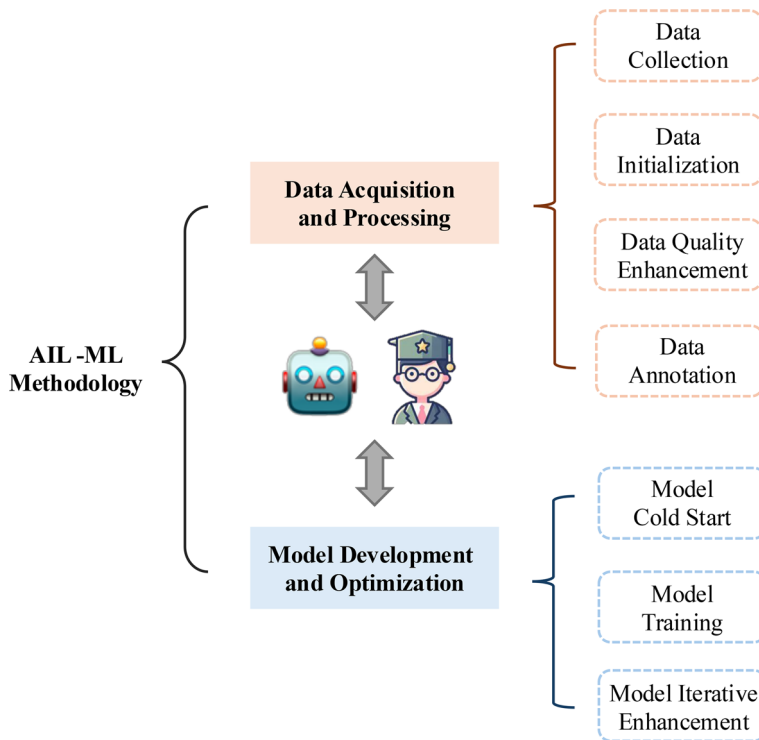


Fig. 2 Framework for AIL-ML Methodology. The framework can be divided into two main phases: Data Acquisition and Processing, and Model Development and Optimization. The first phase contains Data Collection, Initialization, Quality Enhancement, and Annotation. The second phase involves Model Cold Start, Training, and Iterative Enhancement

and optimization. We also summarize applications of AIL-ML in domains requiring specialized knowledge. In the last section, we highlight the current challenges within AIL-ML research and outline future research directions, providing both theoretical and practical guidance for further development in the field.

The insights gained from this paper may contribute to a deeper understanding and practical application of AIL-ML. The paper's contributions are as follows:

1. We conduct a comprehensive literature review of existing works on AIL-ML, focusing on the role of agent in AIL-ML. Specifically, we summarize how humans and LLMs operate within the framework of AIL-ML.
2. We offer a detailed and structured classification of data processing and model development methods within AIL-ML. This taxonomy organizes methods into distinct categories, helping researchers select suitable approaches for specific problems.
3. Based on the classification, we analyze and summarize the differences and relationships between various methods in AIL-ML.
4. We propose a clear methodological framework for AIL-ML, by formalizing a mathematical description. This framework simplifies complex concepts and processes,

enhancing their accessibility to researchers and practitioners. This formalization also establish a solid theoretical foundation for future work.

2 Background

2.1 Related reviews

Human-in-the-loop has emerged as a prominent area of interest within the field of machine learning in recent years. With the expansion of machine learning applications across computer vision, natural language processing, and speech processing, there has been a growing realization that reliance solely on machine-driven learning is limited. Integrating human domain knowledge and experience has proven to be more effective in addressing specific challenges. In the realm of HIL machine learning, several comprehensive review papers have been published, summarizing various techniques, challenges, and future directions within this field. These reviews not only define the various technologies but also discuss the interplay between them, providing researchers with a clear theoretical framework for further exploration. To provide a comprehensive understanding of the state-of-the-art in HIL-ML, we compare three representative surveys that focus on different aspects of this field. Table 1 summarizes their key contexts, theoretical foundations, adaptation to LLMs, strengths, and weaknesses, offering insights into the distinct contributions and limitations of each work.

Wu et al. (2022) conducted a comprehensive overview of HIL-ML from the perspective of data management. They analyzed existing works on HIL, categorizing them into three progressively related categories: (1) efforts to enhance model performance through data processing, (2) improvements in model performance via interventional model training, and (3) human-in-the-loop system design. Furthermore, the paper summarizes applications of HIL and discusses its technical merits and limitations in fields such as natural language processing and computer vision. However, the survey lacks theoretical depth, as it does not formalize the role of human involvement in HIL-ML through mathematical models. Additionally, it does not address the impact of large pre-trained models, which limits its relevance in the context of modern LLMs. Despite these limitations, the survey's focus on practical system implementation offers valuable insights for engineering-focused studies in HIL-ML.

Beyond analyzing HIL from a data perspective, there has been significant research interest in the interaction modalities between humans and machine learning algorithms and leveraging these interactions to boost model performance. Eduardo Mosqueira-Rey et al. have contributed a thorough overview of this area, defining the forms of HIL machine learn-

Table 1 A comparative analysis of surveys on HIL-ML

Dimension	A survey of HIL-ML (Wu et al. 2022)	HIL-ML: state of the art (Mosqueira-Rey et al. 2023a)	Understanding interactions (Cui et al. 2021)	Ours
Covers end-to-end workflow of ML	✓	X	X	✓
Provides a clear theoretical foundation	X	✓	X	✓
Discusses adaptation to LLMs	X	X	X	✓
Emphasizes user interaction and experience optimization	X	X	✓	✓
Shows innovative interaction design	X	X	✓	✓

ing such as Active Learning (AL), Interactive Machine Learning (IML), Machine Teaching (MT), Curriculum Learning (CL), and Explainable AI (XAI) (Mosqueira-Rey et al. 2023a). Their classification hinges on the locus of control within the learning process: active learning with the system in control, interactive machine learning with enhanced user-system interaction, and machine teaching where human experts dictate the learning process. This work provides a delineation of the various interactions and demarcations among different HIL techniques, drawing connections and elucidating their influences. It offers a clear theoretical framework for researchers in the field, laying a foundational theory for the study of HIL. However, it does not address the challenges posed by LLMs, limiting its applicability.

Although the HIL-ML field has explored various interaction types like demonstrations and preferences, there is a notable lack of comparative analysis or guidelines for selecting the most effective type to solve specific learning problems. Yuchen Cui et al. have introduced a set of principles for organizing HIL that examines how different interaction types affect human performance and the quality of training data (Cui et al. 2021). In addition, they discuss strategies for selecting the most effective interaction type for particular learning issues and identify ongoing open questions in the field. While its focus on user-centric design is innovative, the survey covers a narrow scope, lacks concrete application scenarios, and provides minimal theoretical or experimental depth. Furthermore, it does not discuss the potential integration of LLMs with HIL systems, which limits its applicability in addressing contemporary challenges.

While previous reviews have comprehensively addressed the techniques, challenges, and future directions of HIL-ML, the advent of large models has introduced new dynamics and challenges into the HIL-ML landscape. Large models, equipped with extensive prior knowledge, have reduced the reliance on human labor for traditionally labor-intensive tasks such as complex data labeling. Consequently, research in the era of large models has shifted its focus toward using human expertise for tasks like model fine-tuning, output validation, and high-level guidance. This shift has not only lowered the overall cost of HIL systems but also redefined the role of human involvement, making it more strategic and less resource-intensive. To reflect these developments, we reviewed existing works and proposed an updated framework, Agent-in-the-Loop Machine Learning, which integrates human with large-scale models to address the evolving challenges in this domain.

2.2 Related research areas

There are several research fields closely related to AIL-ML, including but not limited to: HIL-ML, Large Models, active learning and knowledge distillation. We briefly describe them in the following.

Human-in-the-loop machine learning (Cui et al. 2021; Wu et al. 2022; Mosqueira-Rey et al. 2023a) is a method that integrates human knowledge to guide and enhance machine learning models. In HIL-ML framework, human participants contribute inputs not only during data preprocessing, model training, and performance evaluation stages but also engage in interactive feedback throughout the entire machine learning workflow. This approach emphasizes the collaborative interaction between humans and algorithms with the goal of optimizing the learning process, thereby improving the accuracy, interpretability, and reliability of the models. HIL-ML is particularly suited for tasks that are too complex for automated systems or require human intuition.

Large model (Brown et al. 2020) typically refers to a type of machine learning model characterized by a substantial number of parameters, often scaling into the billions. These models, also known as large-scale models or foundation models (Bommasani et al. 2021), are designed to process and generate data across various tasks and domains. They leverage deep learning techniques and are predominantly neural network architectures such as transformers (Vaswani 2017), which allow them to achieve state-of-the-art performance in areas like natural language processing and computer vision. Large models are famous for their ability to learn complex patterns and relationships within data, which facilitates advanced reasoning and generalization capabilities beyond specific tasks, contributing significantly to advancements in artificial intelligence.

Active learning (Settles 2009; Mosqueira-Rey et al. 2023a) is a machine learning approach where the learner (often a model) selectively queries an oracle (typically a human annotator acting as a teacher) to label examples that are ambiguous but likely to provide significant insights to the learning process. This targeted approach allows the learner to enhance their performance with fewer training examples. Active Learning is particularly effective in environments rich in unlabeled data where the annotation task is costly or time-consuming. This method strategically reduces the volume of data that needs labeling while maximizing learning efficiency and effectiveness. Active Learning is indeed a method within HIL-ML.

Knowledge distillation (Xu et al. 2024; Gou et al. 2021; Hinton 2015) is a technique where knowledge is transferred from a larger, more complex model (often called the teacher) to a smaller, simpler model (often called the student). The goal of knowledge distillation is to enable the student model to achieve performance comparable to the teacher model but with reduced computational complexity and memory requirements. This is particularly useful for deploying high-performing models on devices with limited resources, such as mobile phones and embedded systems. Knowledge distillation has become a popular method for model compression and is extensively used to enhance the efficiency of deploying deep learning models in resource-constrained environments.

3 Methodology

This section will provide a detailed introduction to the implementation of AIL-ML. As shown in Fig. 3, we categorize the existing work into two main parts: data acquisition and processing, and model development and optimization.

In data acquisition and processing, we discuss several key steps: first, during the data collection stage, we explain how agents participate in and optimize the data collection process (Wu et al. 2020; Hiremath et al. 2022; Hancock et al. 2019; Xu et al. 2022; Long et al. 2023; Ding et al. 2022; Xu et al. 2023; Sahu et al. 2023; Gao et al. 2023; Ye et al. 2022); second, in the data initialization stage, we explore how agents transform collected raw data into formats suitable for machine learning (Zhang et al. 2019; Oh et al. 2019; Kath et al. 2023; Hsieh 2023b; Chen et al. 2021b; Cai et al. 2019; Park et al. 2023); next, during the data quality enhancement stage, we introduce how agents help improve the quality of data (Wu et al. 2020; Xu et al. 2022; Yu et al. 2015; Li 2017, 2017; Yao et al. 2019a; Liu et al. 2021; Wallace et al. 2019; Bartolo et al. 2020; Arakawa et al. 2023; Cho et al. 2023; Wang et al. 2021; Hsieh et al. 2023a; Wang et al. 2023b; Dai et al. 2023; He et al. 2023a; Gao et al. 2023); finally, in the data annotation stage, we explore how agents use their knowledge for

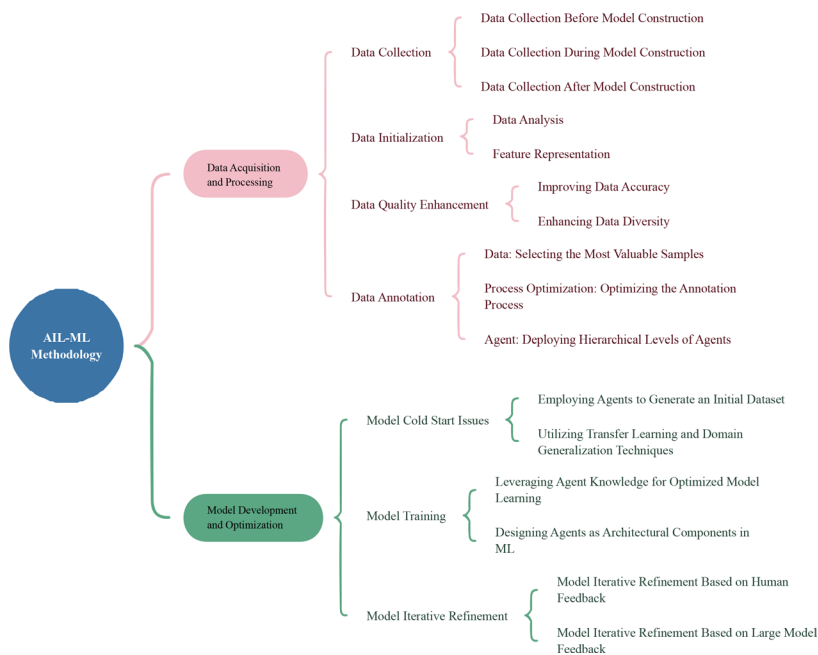


Fig. 3 Taxonomy of Agent-in-the-Loop Machine Learning

precise and effective data labeling (Wu et al. 2020; Xu et al. 2022; Yu et al. 2015; Hancock et al. 2019; Hiremath et al. 2022; Li 2017; Zhang et al. 2019; Liu et al. 2021; Klie et al. 2020; Qian et al. 2020; Kath et al. 2023; Hsieh 2023b; Hemmer et al. 2022; Cui et al. 2021; Koppol et al. 2021; Cai et al. 2019; Wang et al. 2021; Ding et al. 2022; He et al. 2023b; Hsieh et al. 2023a; Sahu et al. 2023; Wang et al. 2024; Lu et al. 2024).

In the model development and optimization part, the focus is on three core segments: firstly, we discuss how AIL-ML strategies effectively address the model cold start problem (Hiremath et al. 2022; Zhang et al. 2019; Wang et al. 2022; Ben-David et al. 2006; Zhuang et al. 2020; Xu et al. 2022; Arakawa et al. 2023; Wang et al. 2021; Ding et al. 2022; Xu et al. 2023; Hsieh et al. 2023a; Ye et al. 2022; Li et al. 2024); secondly, during the model training phase, we detail how agents' advanced knowledge is used to calibrate model parameters and optimize the learning framework (Arous et al. 2021; Oh et al. 2019; Kath et al. 2023; Wei et al. 2022; Wang et al. 2023a; Roels et al. 2019; Weber et al. 2020; Kwon and Michael 2023); and lastly, in the model iterative enhancement phase, we emphasize how continual intervention and feedback from agents incrementally enhance the performance of machine learning models through multiple iterations (Wu et al. 2020; Hancock et al. 2019; Yao et al. 2019b; Qian et al. 2020; Stiennon et al. 2020; Ouyang et al. 2022; Crochepierre et al. 2022; Fan et al. 2024; Ahn et al. 2023; Arakawa et al. 2023; Ziegler et al. 2019; Xu et al. 2023; He et al. 2023a; Sun et al. 2024; Klissarov et al. 2023; Guo et al. 2024).

By introducing the AIL-ML concept, we effectively leverage the strengths of human and large models to construct efficient, accurate, and adaptable machine learning models. Agents play a pivotal role in data acquisition and processing, as well as in model development and optimization. This strategy of combining human wisdom with machine power

offers a new possibility for constructing efficient, precise machine learning models, pushing the technology to higher levels of achievement.

3.1 Data acquisition and processing

In the field of machine learning, data and models are intricately linked. The collection and initialization of data directly influence the construction and performance of models. Moreover, enhancing data quality is crucial for the iterative improvement of models. This interaction between data and models forms a mutually beneficial relationship, highlighting the importance of data quality in improving model performance. In this section, we investigate AIL-ML techniques that focus on optimizing the entire data lifecycle in machine learning. This includes a detailed examination of methods in the areas of *Data Collection*, *Data Initialization*, *Data Quality Enhancement*, and *Data Annotation*.

3.1.1 Data collection

In this section, we explore the data collection mechanisms within the AIL-ML framework, where agents can be humans, large models, or a combination of both. Agents play a crucial role in generating diverse, comprehensive, and representative data sets, essential for training complex machine learning models. This not only broadens the scope of training data but also enhances the model's generalization capabilities and adaptability to complex environments.

3.1.1.1 Mathematical framework for data collection In this part, we establish a foundational mathematical framework to systematically describe the data collection mechanisms within the AIL-ML architecture. This framework is summarized by the following general equations, which form the various data collection strategies discussed in subsequent sections.

$$x_t = f(S_t, D_t, M_t, A_t) \quad (1)$$

where:

- x_t represents the data generated by the agent at time t ,
- S_t represents the seed datasets at time t ,
- D_t and M_t represent the current dataset and model, respectively,
- A_t denotes the actions or decisions of the agent at time t , which may include inputs from humans H_t and large models L_t ,
- f is a function defining how data is generated based on the current data set, model state, agent state, and actions.

Additionally, the process by which agents generate data based on given labels can be formalized through a conditional probability model:

$$x_t = p(x \mid y, S_t, D_t, M_t, A_t) \quad (2)$$

where:

- x_t is the data generated by the agent at time t ,
- y is the provided label,
- $p(x \mid y, S_t, D_t, M_t, A_t)$ is a conditional probability distribution indicating the likelihood of generating data x given the label y , current seed dataset S_t , data set D_t , model state M_t , and agent actions A_t .

These formulations offer a comprehensive description of data collection under the AIL-ML framework, illustrating how to utilize agents' internal state, model state, and available data, along with potential external labels, to acquire data. This highlights the dynamic and adaptive nature of AIL-ML. In the subsequent sections of the chapter, each data collection strategy will be mapped to these general formulas.

Research analysis indicates that the AIL-ML framework utilizes three primary data collection strategies, each corresponding to different stages of the machine model's lifecycle. These strategies include data collection before model construction, during model construction, and after model construction.

3.1.1.2 Data collection before model construction Before model construction, data collection is suitable for the following two scenarios. Firstly, these strategies are suitable for tasks that require highly specific data and lack large-scale public datasets (Long et al. 2023). For example, in the Internet of Things (IoT) context, various sensors are positioned at different locations. Due to the lack of public dataset, researchers need to collect the data. It is essential that the data should be not only abundant but also diverse and representative, to effectively capture the diverse potential states of the real-world environment. For instance, Hiremath et al. (2022) demonstrate how data is passively observed and collected through sensor networks installed in different locations within smart home systems. This network, containing door sensors, motion sensors, and temperature sensors, continuously monitors environmental changes and logs events with timestamps and sensor IDs. The data, stored in event logs on local servers or in the cloud, forms the basis for subsequent processing and analysis. Similarly, Wu et al. (2020) presented a system that automatically collects environmental sound signals through smart speakers. This system utilizes unsupervised clustering techniques to identify recurring activity types. Once the system can identify specific activities with high confidence, it starts a one-shot interaction with users through voice commands. This interaction is used to label these activities and create a labeled dataset. This approach allows the system to improve the accuracy of environmental sound recognition incrementally through Continuous learning and in-situ training, without the need for predefined data. Consequently, it develops an accurate, environment-specific, acoustic signal-based activity recognition model. In this scenario, the data collection process can be modeled as:

$$x_t = f(\emptyset, \emptyset, \emptyset, A_t) \quad (3)$$

Here, $\emptyset$ indicates that the seed dataset S_t , the current dataset D_t and the model state M_t are not involved in the data generation process in this data collection strategy. This is due to the lack of this information before the model has been constructed. For convenience in expression throughout the remainder of this paper, $\emptyset$ will not be explained any more. A_t represents the user's daily activities at time t . In this case, A_t represents the user's daily

activities, e.g., daily operations in the smart home, which generate corresponding data x_t through sensor activation or sound recording. The function f defines how the sensor data is collected and generated based on the behavior A_t of the agent through sensor activations or sound recordings.

Secondly, data collection before model construction is suitable for tasks leveraging large models for data generation. Trained on vast amounts of data, large models have accumulated extensive general knowledge and demonstrated remarkable abilities in reasoning and analysis. This allows researchers to use large models to generate high-quality data, significantly reducing the time and costs. This enables both individuals and small organizations to utilize advanced AI technologies more effectively. Through well-designed prompt engineering, researchers can direct large models to generate specific datasets on demand, optimizing both the training process and model performance. Ding et al. (2022) proposed an approach named Prompt-Guided Training Data Generation (PGDG). This method leverages the generative capabilities of GPT-3 to produce labeled data pairs directly from specific prompts, significantly enhancing the efficiency and quality of data generation. For tasks such as sentiment analysis and relationship extraction, the data generated through this method matches or even surpasses the quality of manually annotated data with reducing time and cost. Additionally, large models can generate diverse data using seed datasets. In Xu et al. (2023), C Xu et al. detailed how to use seed datasets to collect high-quality, multi-turn dialogue data through self-chat. They chose platforms like Quora and Stack Overflow as seed datasets. These platforms provide a wealth of user-generated questions that guide the generation of dialogues by ChatGPT, which simulates both sides of a conversation. Each dialogue began with a seed question and naturally progressed through several rounds of exchanges until it ended. To ensure the quality and format of the dialogues, researchers used predefined templates to control the dialogue generation process. This method not only addresses the scarcity of high-quality, multi-turn dialogue but also offers a practical solution due to its efficiency and relatively low cost. To further enhance the diversity and quality of the generated data, researchers have designed various generation strategies (Ye et al. 2022; Gao et al. 2023; Sahu et al. 2023), such as Greedy Search, Top-k Sampling, and Nucleus Sampling. These strategies aim to improve the diversity and quality of the generated text. Such innovative techniques ensure that the data generated is both varied and of high quality, suitable for training robust models. This data generation process can be described by the equation:

$$x_t = f(S_t, \emptyset, \emptyset, A_t) \quad (4)$$

where S_t represents the seed dataset, such as user questions from Quora and Stack Overflow in Xu et al. (2023). A_t denotes the agent's actions, such as ChatGPT generating dialogues based on the seed questions. The function f defines a function or process that generates data x_t based on S_t and A_t .

3.1.1.3 Data collection during model construction During model construction, the continuous collection of data through agents can enable the model to better adapt and learn under complex or unfamiliar environmental conditions. Moreover, this can make model identify potential biases. In Hancock et al. (2019), data collection spans the entire lifecycle of the dialogue model, particularly post-deployment. Through real-time interactions with users, the system continuously collects new training data, including Human-Bot (HB) dialogue

data and user feedback data. When the model assesses high user satisfaction in a conversation, the user's responses are recorded as HB dialogue data. Correspondingly, when the model anticipates potential errors, it requests feedback from users, which is then used to refine the model. This collected data, after processing and storage, is regularly utilized for retraining the model, enabling it to continually learn and improve in practical applications, thereby enhancing conversational abilities and user satisfaction. This approach not only reduces the dependency on manually annotated data but also allows the model to rapidly adapt to new conversational contexts, significantly improving user experience and model performance. The data collection process in this scenario can be represented by:

$$x_t = f(\emptyset, \emptyset, M_t, A_t) \quad (5)$$

In this context, M_t represents the state of the model at time t , such as the chatbot in Hancock et al. (2019). The state of the model evolves over time as more training data is accumulated and the model is updated. A_t denotes the agent's actions, which include user interactions and feedback that generate new training data. And f is the core mapping function of the data collection process. In this context, it transfers the current model state M_t and agent's action A_t to new training data x_t .

3.1.1.4 Data collection after model construction After model construction, the collection of data from agents can allow models to handle new types of tasks. For example, in gesture recognition systems, allowing users to personalize their gestures can enhance gesture memorability, increase interaction efficiency, and improve accessibility for individuals with specific needs. Xu et al. propose a framework for gesture customization using few-shot learning, where users provide only a few custom gesture examples to the system (Xu et al. 2022). This method enables the system to recognize customized gestures with high accuracy without compromising the performance of existing gestures. Specifically, users record new gestures on a smartwatch interface, and the system guides them through data collection via an interactive interface. The system continuously monitors performance and provides feedback to users. If the model's accuracy falls below expectations, the system guides users to record more samples to further optimize the model. This data collection process can also be described by the equation:

$$x_t = f(S_t, D_t, M_t, A_t) \quad (6)$$

where S_t is the seed dataset, such as negative samples (e.g., daily activities) and existing gestures. D_t represents the training dataset at time t , which evolves as new customized gestures in Xu et al. 2022. M_t denotes the current model such as the gesture recognition model in Xu et al. 2022. And A_t stands for the agent's actions, such as providing customized gesture data to the system in Xu et al. 2022. Through S_t , D_t and M_t , the system can prevent the agent's custom gestures from being too similar to existing gestures, thereby avoiding overfitting to new gestures and forgetting old gestures. Additionally, by incorporating negative samples in S_t , the system can prevent conflicts between new gestures and the user's daily activities.

3.1.1.5 Analysis of data collection in AIL-ML In the aforementioned sections, we have provided mathematical representations for three distinct data collection strategies corresponding to different stages of the model lifecycle: before model construction, during model construction, and after model construction. These mathematical formulations vary in their functional structures and conditional elements, reflecting the unique data collection methodologies and technical characteristics inherent to each phase. By comparing these mathematical models, we can discern the specific features and limitations of each strategy, understand the trade-offs they entail, and identify the conditions under which the formulas might be extended to yield additional benefits.

Data Collection Before Model Construction primarily focuses on assembling comprehensive and representative datasets necessary for initializing the model. At this stage, the data collection process is typically represented by the equations $x_t = f(A_t)$ or $x_t = f(S_t, A_t)$. Prior to model construction, researchers generally lack a trained model or sufficient existing data, rendering M_t and D_t either nonexistent or minimal. Consequently, data collection is primarily driven by external factors such as sensor monitoring, user activities, or data generated by large models based on seed datasets. The simplified function $f(A_t)$ or $f(S_t, A_t)$ reflects an externally driven data collection process, where user activities or prompts guide data generation without considering the model's dynamic adaptation. This omission of M_t reduces the complexity but means that data quality and diversity rely only on external conditions like activity patterns and the scope of seed datasets, lacking a feedback loop from the model. This approach offers the advantage of accumulating substantial data prior to model training, reducing potential data gaps in later stages, and leveraging large models to enhance data quality while minimizing annotation costs. However, the main challenge lies in designing effective data collection strategies and avoiding errors or biases in the data generated by large models. The absence of M_t leads to a more efficient and straightforward data generation process, but it sacrifices the ability to collect data tailored to the model's evolving needs, highlighting the trade-off between simplifying data collection at the expense of adaptive data collection capabilities.

Data Collection During Model Construction integrates data acquisition into the training process itself. During this phase, the data collection process is typically modeled as $x_t = f(M_t, A_t)$. Unlike the pre-construction stage, the model's current state M_t plays a significant role in guiding data collection. This phase utilizes real-time data generated through user interactions and user feedback. The model is in a transitional state, undergoing training or initial deployment with partial parameters and structures. The collected data reflects actual usage scenarios, facilitating rapid model adaptation and continuous optimization. Incorporating M_t into the function allows data collection to be dynamically adjusted based on the model's current performance and identified biases, facilitating continuous improvement and adaptation. The key benefits include data that closely reflects deployment environments, enabling rapid iterations and mitigating bias, while also reducing dependence on manual annotations. However, challenges related to data noise and variability in feedback quality remain, requiring robust feedback processing and ongoing model maintenance. Including M_t enhances data collection specificity and adaptability but introduces additional complexity. Real-time access to and processing of the model's state require more advanced infrastructure, increasing both computational and operational overhead.

Data Collection After Model Construction emphasizes the personalization and extension of the model's capabilities to handle new tasks. In the post-construction phase, data collection is represented by $x_t = f(D_t, M_t, A_t)$, sometimes including S_t . This comprehensive formulation accounts for the existing dataset (D_t), the model's state (M_t), and the agent's actions (A_t). With a fully trained and deployed model, data collection focuses on personalization and task extension. The existing dataset ensures consistency and prevents the model from forgetting previously learned information, while the model's state guides the acquisition of new, relevant data. By integrating D_t and M_t , the system can effectively manage the introduction of new data, ensuring that user-specific data do not overlap excessively with existing data, thereby avoiding overfitting and maintaining overall model performance. Data collection mechanisms here are user-driven, with agents actively providing specific data samples based on performance feedback. This strategy allows for the addition of new tasks without degrading existing model performance by preventing overfitting to new data and forgetting previously learned information through the inclusion of negative samples. However, challenges include avoiding excessive bias towards individual users and ensuring a balanced integration of new data with existing knowledge.

The equation $x_t = f(S_t, D_t, M_t, A_t)$ can flexibly represent various data collection scenarios by selectively omitting certain terms based on the specific phase. This flexibility allows the framework to adapt to different strategies without compromising its general applicability. Introducing additional terms such as M_t or D_t in specific contexts enables finer control and optimization of data collection strategies. For example, evolving from $f(A_t)$ to $f(M_t, A_t)$ allows for model-driven data collection, enhancing iterative training and correction processes. From the analysis, we can derive the following high-level insights:

- Data collection strategies can align with the model's lifecycle stages. Before model construction, extensive data accumulation and generation are crucial for robust initial training. During construction, real-time data collection supports continuous learning and bias correction. After construction, personalized data collection enables the model to adapt to specific user needs and new tasks.
- Each phase involves trade-offs. Pre-construction strategies reduce annotation costs and ensure data diversity but require careful planning and quality control. During construction, real-time data accurately captures user interactions but introduces variability and noise. Post-construction strategies enhance personalization and adaptability but risk overfitting and require effective data management to preserve existing knowledge.
- Methodological versatility is essential, enabling researchers to select or combine strategies based on task requirements, data availability, and resource constraints. This ensures data quality, diversity, and dynamic adaptability within the AIL-ML framework.

3.1.2 Data initialization and preprocessing

Data initialization and preprocessing are crucial steps that transform raw data into a format suitable for machine learning models. In this section, we will introduce the AIL-ML techniques that initialize and preprocess data through agents.

3.1.2.1 Mathematical framework for data initialization and preprocessing How to utilize agents for data initialization and preprocessing can be described as follows:

$$F_t = \psi(D_t, A_t, P_f) \quad (7)$$

where:

- F_t represents the feature set derived at time t ,
- as defined previously, D_t and A_t continue to denote the current data set and the agent's actions or decisions,
- function ψ serves as a data transformation function, incorporating all essential preprocessing steps such as data cleaning, normalization, and feature extraction,
- P_f stands for the set of preprocessing parameters.

Upon reviewing AIL-ML literature, we identified two main methodologies that utilize agents for data initialization and preprocessing: Data Analysis and Feature Representation. These methodologies leverage agents to perform data analysis (Zhang et al. 2019; Hsieh 2023b; Chen et al. 2021b), Wei et al. 2022 or to develop effective feature representation (Oh et al. 2019; Kath et al. 2023; Cai et al. 2019).

3.1.2.2 Data analysis Data analysis by agents involves examining large datasets to discern underlying patterns and valuable insights that are not immediately apparent. This process is vital for preparing data in a way that significantly enhances the training and performance of machine learning models. Zhang et al. (2019) employed a high-recall regular expression to detect potential entity candidates within texts. Initially, experts design preliminary regular expressions suitable for specific entity extraction tasks, such as dates, email addresses, and course numbers, aiming to capture most target entities. Then, these expressions are used to scan documents, extracting all matching substrings. Each matched substring is expanded to include surrounding text, providing sufficient contextual information for its validation as a target entity. This method creates a collection of candidate substrings, encompassing all potential entities, significantly narrowing the scope of text that needs further processing and thereby enhancing the efficiency of subsequent annotation and model training. For tasks that require expert knowledge, the AIL-ML strategy distills the deep knowledge and understanding of human experts into ML models through data analysis. Hsieh (2023b) leveraged expert knowledge in the data initialization and preprocessing stages. They utilize chest X-ray images, eye-tracking data from radiologists, and patients' clinical data. The eye-tracking data, which includes gaze points and sequences during the diagnostic process, is mapped to image coordinates to extract the visual search patterns of the radiologists. Concurrently, clinical data is encoded and standardized to align dimensionally with 3D image data, and expanded through convolution layers for effective integration with image data. Through these steps, the diagnostic behaviors and knowledge of human experts are effectively distilled into the model, enabling it to learn from the experts' diagnostic processes.

In this scenario, the data initialization and preprocessing can be formalized by Eq. 7. The specific components of the formula are summarized in Table 2.

Table 2 Symbol descriptions for data analysis

Symbol	HIL entity extraction (Zhang et al. 2019)	Chest X-Ray diagnosis (Hsieh 2023b)
D_t	Text dataset	Chest X-ray images
A_t	Design of regular expressions for entity extraction	Looking at the chest X-ray images during the diagnostic process
P_f	Time allocation between constructing regular expressions and labeling data	Parameters for data encoding and integration

3.1.2.3 Feature representation On the other hand, agents are used for feature representation, which transforms raw data into a structured format more suitable for subsequent algorithmic processing. This transformation ensures that the data are formatted to maximize the efficiency and accuracy of the machine learning model. In Oh et al. (2019), agents' interaction and feedback play important roles in feature representation. Users annotate video frames, for instance by delineating foreground and background, providing essential feature information. These annotations are used to generate object masks and serve as feature inputs to help the network and represent foreground objects. The user-provided data is then encoded into a format suitable for network processing, such as binary masks representing positive and negative examples. The interaction network combines this information with video frame data and transforms it through a feature encoding module into high-dimensional features suitable for model processing. Ultimately, these encoded data are integrated into a multi-channel input tensor for the interaction and propagation networks, ensuring the model effectively utilizes the initial information provided by users to enhance performance and prediction accuracy. Beyond low-dimensional data, agents can also represent feature for high-dimensional sensory data, such as images. Park et al. (2023) proposed Localized Symbolic Knowledge Distillation (LSKD), a framework designed for efficient image feature representation. This framework leverages multi-modal large models to extract global image features, encompassing scenes, objects, and conceptual information, which are organized into Global Descriptors. Simultaneously, candidate regions within the image are identified, and fine-grained Local Descriptors are generated for each region using vision-language models. Additionally, LLMs generate commonsense knowledge tied to specific regions, which is refined through a supervised "critic model" to remove inconsistencies. This approach effectively integrates global and local information. It also enhances semantic representations through dynamic reasoning, providing robust inputs for subsequent inference and learning tasks.

Furthermore, agents can optimize feature representation. Kath et al. (2023) presented an interactive machine learning tool for annotating passive acoustic monitoring datasets created for wildlife monitoring. Users can select data points within a state-space representation and create new boundaries, further refining feature representation. Through real-time feedback, the model dynamically adjusts feature representations and prediction outcomes, gradually improving accuracy with each retraining cycle. Cai et al. (2019) developed tools that helped pathologists cope with the shortcomings of deep learning algorithms during image retrieval. These tools are designed to adjust retrieval algorithms in real time to better meet the diagnostic needs of doctors and enhance their trust in the algorithms. While using the SMILY system, pathologists refine and optimize image feature representations

through tools like regional, example-based, and conceptual refinement. These tools allow pathologists to emphasize or remove irrelevant features, ensuring the model focuses on correct diagnostic features. This process of real-time interaction and feedback can be seen as a dynamic method of data preprocessing, aiding in the improvement of the model's feature representation.

This kind of data initialization and preprocessing can be described by Eq. 7. A detailed listing of the parameters involved is provided in Table 3

3.1.2.4 Analysis of data initialization and preprocessing in AIL-ML We analyze two primary methodologies of data initialization and preprocessing in the AIL-ML literature: Data Analysis and Feature Representation.

Data Analysis is to derive richer context and insights from raw data. Agents assist in identifying critical patterns, candidate entities, and key structural elements that may not be easy to detect. For example, through expert-driven strategies such as hand-crafted regular expressions or the integration of human-derived signals (e.g., eye-tracking data from radiologists), data analysis steps ensure that downstream ML models start from a more semantically meaningful and contextually informed baseline. Human expertise often guides data analysis through the design of rules or domain-specific entity extraction patterns. Agents can leverage human guidance to more efficiently sift through large datasets, reducing noise and focusing attention on key data segments.

Feature Representation focus on how to encode that information so it becomes directly usable by machine learning models. Agents and users collaboratively shape the feature space to highlight the most relevant characteristics of the data-refining object masks in video frames, delineating meaningful acoustic signals from background noise, or emphasizing clinically significant image features. These approaches transfer high-dimensional data into representations that enhance learning performance. Agent feedback can be real-time and iterative, allowing agents to refine, reshape, and optimize the feature space as the model trains.

3.1.3 Data quality enhancement

Data quality is important for model training and its subsequent performance. In this section, we explore AIL-ML methods related to enhance data quality. By identifying and correcting discrepancies, biases, and errors in raw data, agents significantly enhance the precision and

Table 3 Symbol descriptions for feature representation

Symbol	User-guided video segmentation (Oh et al. 2019)	HIL acoustic annotation (Hsieh 2023b) and Human-centered medical tools (Cai et al. 2019)	LSKD (Park et al. 2023)
D_t	Video frames	Raw data	Images
A_t	User annotations indicating foreground and background regions, directly influencing the feature extraction process	Real-time user feedback and interactions that guide feature refinement	Generating global and local descriptions
P_f	Parameters related to mask generation	Parameters for feature optimization	Parameters for feature extraction

consistency of the data. Moreover, agents play a key role in augmenting the diversity and richness of data

3.1.3.1 Mathematical framework for data quality enhancement To explain how agents enhance data quality within AIL-ML framework, we adopt the following general formula:

$$D_{t+1} = \theta(D_t, A_t, Q) \quad (8)$$

where:

- D_{t+1} denotes the enhanced dataset at time $t + 1$ after data quality enhancement,
- as defined previously, D_t and A_t continue to denote the current data set and the agent's actions or decisions,
- Q represents a set of parameters for data quality enhancement, defining specific quality control operations (e.g., noise thresholds, labeling consistency checks, or optimization strategies),
- θ represents a general-purpose function for enhancing data quality through a series of operations. These operations include data cleaning, data diversity enhancement by synthesizing new samples and expanding the dataset, improving consistency and accuracy by adjusting mislabeled samples and reducing uncertainty and so on.

There are two primary AIL methodologies by which agents can enhance data quality: 1) improving data accuracy or consistency, (Wu et al. 2020; Yu et al. 2015; Li 2017; Arous et al. 2021; Arakawa et al. 2023; Cho et al. 2023; Gao et al. 2023, 2024) and 2) increasing data diversity (Xu et al. 2022; Yao et al. 2019a; Liu et al. 2021; Wallace et al. 2019; Bartolo et al. 2020; Hsieh et al. 2023a; Dai et al. 2023).

3.1.3.2 Improving data accuracy The first methodology focuses on improving the accuracy and consistency of the data. For machine learning, enhancing data precision and consistency can improve model performance. The knowledge and analytical capabilities of agents can significantly augment data accuracy, thereby enhancing model precision. In the context of smart home environments, Cho et al. (2023) introduced a novel strategy named AI-to-Human (AHA), which actively modifies environmental conditions to enhance the inferential accuracy of AI models. Unlike traditional smart devices, such as embedded vision AI sensors that passively monitor resident activities, AHA actively induces resident behaviors, facilitating more reliable AI inferences. Initially, AI sensors detect poor perception conditions, such as orientations away from the camera or distances too close or too far from the camera. Then, the AHA system employs smart home devices like speakers, displays, or ambient lighting to unremarkably guide resident behavior into more favorable perceptual conditions. The AHA system effectively improves data quality and overcomes the common challenges of passive perception inherent in traditional methods. Similarly, Listen Learner (Wu et al. 2020) employs an active learning approach that enhances data quality through various interactive strategies with users, including open-ended questions, confirmation questions, and refinement questions. When a new event falls into the decision boundary of the model, the system will ask a choice question as a confirmation question. This not only

reduces the cognitive load on users but also enhances the quality of data, adapting dynamically to different environmental sounds, thus improving data precision and the accuracy of classification models. Furthermore, some AIL-ML methods enhance data precision by incorporating human reasoning. For instance, the MARTA framework (Arous et al. 2021) introduces human rationales combined with Bayesian methods to assess agent reliability, thereby enhancing data precision quality. Agents provide not only document labels but also rational segments supporting these labels. During training, the model weights these rational segments based on agent reliability, ensuring that high-quality rational segments contribute more.

Researchers have implemented various methods to manage and control the quality of agent annotations. In LSUN (Yu et al. 2015), annotation quality control is achieved through several strategies including redundant annotations, quality checks, and detailed annotation guidelines. Specifically, each item is annotated by at least two independent annotators to ensure consistency in the results. Items with known labels are embedded within the annotation tasks to monitor the annotators' performance. Both online and offline quality control mechanisms ensure that annotators deliver high-quality results. Detailed annotation guidelines and examples help annotators understand the requirements of the task clearly. Together, these methods ensure the high quality and consistency of the annotated data.

Some researchers employed agents to identify and correct discrepancies, biases, and errors in raw data, significantly enhancing data consistency and model accuracy. Gao et al. (2024) proposed LLMIE-UHAR, which leverages Large Language Models (LLMs) and Iterative Evolution for Unsupervised Human Activity Recognition. This method capitalizes on the logical reasoning capabilities of LLMs to detect inconsistencies within datasets. This is particularly important for unsupervised learning where labels are absent. The effectiveness of LLMIE-UHAR shows the potential of LLMs to improve data quality in real-world applications. This capability of agents to augment data quality forms a critical component in the development of more robust, efficient, and reliable machine learning systems.

There are some AIL methods that enhance data precision by reducing data noise. Arakawa et al. (2023) proposed PrISM-Tracker, which incorporates human input to manage data uncertainty. This framework actively interacts with user input when model predictions are uncertain, effectively reducing noise and uncertainty of data. Particularly in smart home environments, where sensor data may be noisy or the sequences of actions unclear. Additionally, to address the challenge of initial low-quality data generation by large models, Gao et al. (2023) introduced the SUNGEN framework, which uses language models to generate synthetic data with task descriptions and label information. This initial data often includes samples with incorrect labels or irrelevant content that can lead to model overfitting and reduced generalization. The SUNGEN framework fixes this by employing a dual-layer optimization process: an inner loop optimizes model parameters using weighted cross-entropy loss, while an outer loop uses a noise-robust loss function to evaluate model performance on a synthetic validation set. This iterative process refines sample weights, enhancing the selection of high-quality data.

To make it clearer how the cited works integrate into Eq. 8, we present them in Table 4.

3.1.3.3 Enhancing data diversity The second methodology emphasizes broadening the range of data variability, which is vital for training robust models capable of performing

Table 4 Symbol description for methods emphasizing data accuracy and consistency

Method	Agent actions (A_t)	Key parameters (Q)	Data quality enhancement function (θ)	Output (D_{t+1})
AHA (Cho et al. 2023)	Modifying environment (e.g., lighting or orientation) to improve perception	Perception thresholds, environment triggers	Detecting poor conditions and adjusts environment	More accurate sensor data
Listen Learner (Wu et al. 2020)	Asking confirmation or refinement questions to users for uncertain events	Uncertainty bounds, question formats	Identifying uncertain samples and refining labels via user feedback	Refined dataset with reduced boundary noise
MARTA (Arous et al. 2021)	Providing human rationales along with labels	Bayesian weighting parameters	Weighing agent reliability and rationales to adjust final labels	Refined labeling in D_{t+1} , emphasizing high-quality rationales and improving overall data consistency
LSUN (Yu et al. 2015)	Providing multiple labels, with known-label checks as a control	Required annotations per item, known-label embedding, online/offline QC thresholds	Merging annotator inputs and filters low-quality labels	Consistent and high-quality data, reduced annotator errors
LLMIE-UHAR (Gao et al. 2024)	Detecting and correcting dataset inconsistencies	LLM prompts, logical constraints	Using LLM reasoning to identify mismatched samples and correct them	Logically consistent data
PrISM-Tracker (Arakawa et al. 2023)	Correcting the data when model predictions are uncertain	Uncertainty triggers, user feedback rules	Collects user clarifications to disambiguate noisy sensor data	Reduced noise and uncertainty in sensor readings, better clarity in time-series
SUNGEN (Gao et al. 2023)	Employs a dual-layer optimization for synthetic data	Weighted cross-entropy, noise-robust loss	Iteratively adjusts sample weights and filters out faulty synthetics	Higher overall data quality, fewer mislabeled synthetic samples

well across diverse scenarios. Enhancing the diversity of data can not only improve the generalization ability but also ensure superior performance on unseen samples of machine learning models.

Synthetic data generation is a prevalent method for enhancing data diversity in machine learning. This approach involves creating new data based on the characteristics of existing data to increase the diversity of the data available for model training. Liu et al. (2021) introduced a method for global feature contribution analysis, which combined human evaluation to identify error features of the model. This method performs a local feature contribution analysis and quantifies each feature's contribution. Subsequently, it calculates the global contributions of each feature to identify those with significant impacts on model predictions. Agents are employed to annotate and verify these global features, evaluating their contribution and correctness. This methodology not only improves the generalization capabilities of the data but also model's performance on unseen data, enhancing the robustness and reliability of model's. Moreover, large models can synthesize data to increase diversity. In Hsieh et al. (2023a), after employing Chain-of-Thought prompting techniques, large models

generate natural language inferences that include both the final labels and detailed intermediary steps and logical explanations, providing richer and more useful training data for smaller models. The inferential content generated by large models significantly improves the context and detail quality of the training data.

By adding small perturbations to the original samples, adversarial examples can greatly reduce the accuracy of the original classifier and achieve the purpose of anti-deep learning Pan et al. (2020). Because past approaches expose superficial patterns, the resulting adversarial examples are limited in complexity and diversity. To address these shortcomings, Wallace et al. (2019) proposed an AIL-ML framework for generating adversarial examples. Researchers have applied this generation framework to a question answering task called Quizbowl, where trivia enthusiasts craft adversarial questions. The questions generated within this framework were tested through live human-computer matches, demonstrating that while these questions appeared ordinary to human participants, they systematically stump neural and information retrieval models. Agent-involved adversarial example generation is effective, which not only increases the diversity and complexity of the data but also enhances the models' generalization capabilities (Bartolo et al. 2020).

Data augmentation plays a pivotal role in enhancing data diversity. Xu et al. (2022) have developed a user-defined gesture recognition framework that requires participants to provide a few custom gesture examples. After acquiring these gestures, the system employs multiple data augmentation methods to generate additional samples: 1) zooming, to simulate different gesture speeds; scaling, to simulate different gesture strengths; time-warping, to simulate gesture temporal variance. Thus, these augmented samples increase the diversity of the data. Additionally, large models, pre-trained on extensive datasets, have been fine-tuned with reinforcement learning from human feedback (RLHF) to generate text similar to human expression. Dai et al. (2023) utilized large models to generate diverse and high-quality textual data as data augmentation, which rephrases each sentence in the training samples into multiple conceptually similar but semantically different samples. The augmented samples can then be used in downstream model training. Data augmented through large models significantly boosts performance in few-shot learning tasks. This efficient and practical data augmentation method reduces reliance on manual annotation. This data augmentation method is efficient and practical, reducing reliance on manual annotations.

These strategies demonstrate how researchers utilize AIL methods to realize data in data quality enhancement for machine learning, ensuring that models are not only trained on high-quality data but also adaptable to complex real-world environments.

Similarly, to clarify how these diversity-oriented approaches align with Eq. 8, we present them in Table 5.

3.1.3.4 Analysis of data quality enhancement in AIL-ML As discussed above, we have introduced mathematical representations for two primary data quality enhancement methodologies within the AIL-ML framework: Improving Data Accuracy and Consistency and Enhancing Data Diversity. These methodologies correspond to the Eq. (8). By comparing these methodologies across various aspects, we can gain a comprehensive understanding

Table 5 Symbol description for methods emphasizing data diversity

Method	Agent actions (A_t)	Key parameters (Q)	Data quality enhancement function (θ)	Output (D_{t+1})
HIL error detection (Liu et al. 2021)	Annotating and verifying global error features	Feature contribution thresholds, annotation rules, error detection criteria	Refining data by combining feature contribution analysis with human evaluation	A dataset with identified error features addressed, yielding better generalization and robustness
Step-by-step distillation (Hsieh et al. 2023a)	Generating detailed reasoning steps	Prompt design	Incorporates reasoning and explanations into the original data	A refined dataset with enriched contextual
AIL adversarial examples (Wallace et al. 2019; Bartolo et al. 2020)	Crafting adversarial questions or inputs that intentionally confuse models	human interaction protocols, adversarial crafting strategies	Generating adversarial data to diversify training data	A dataset enriched with diverse and complex adversarial examples, improving model robustness and generalization
Gesture augmentation (Xu et al. 2022)	Providing a few custom gestures from users, which are then augmented	Zoom factors, scaling intensity, time-warp parameters	Applying various transformations (zoom, scale, warp) to user-defined gestures	A gesture dataset covering diverse speeds and strengths
Augpt (Dai et al. 2023)	Utilizing large models to rephrase or expand textual samples	RLHF fine-tuning configs, sampling settings	Using large models to generate multiple conceptually similar but semantically different variations of each sentence	A diversified text corpus with semantically varied samples

of their distinct roles, advantages, and limitations in enhancing data quality for machine learning models.

Improving Data Accuracy and Consistency focuses on enhancing the precision and reliability of the existing dataset by addressing errors, noise, and inconsistencies. This approach

ensures that the data used for model training is accurate, thereby directly contributing to improved model performance. These methods can be detailed as follows:

- **Correcting Mislabelled Samples:** Utilizing agnet to rectify labeling errors.
- **Label Refinement:** Incorporating human rationales to enhance label quality.
- **Consistency Checks:** Implementing quality control mechanisms to ensure data consistency.

These methods rely on quality control parameters, such as label accuracy thresholds, agent reliability metrics, and detailed annotation guidelines, to maintain high standards of data quality. Consequently, the enhanced dataset D_{t+1} is characterized by higher accuracy, reduced noise, and greater consistency, which will enhance the model's precision and reliability.

Enhancing Data Diversity aims to increase the variability and richness of the dataset to improve the model's ability to generalize across diverse scenarios. This methodology introduces new data samples or transforms existing ones to create a more varied dataset, thereby making the model more robust and adaptable. The specific strategies implemented include:

- **Data Augmentation:** Applying techniques such as scaling, time-warping, or zooming to generate additional samples from existing data.
- **Adversarial Example Generation:** Creating challenging inputs designed to test and enhance model robustness.
- **Synthetic Data Generation:** Utilizing agent to produce diverse and contextually rich training data.

These methods depend on specific augmentation parameters, perturbation strategies, and prompt designs to ensure that the generated data effectively increases diversity while maintaining quality. As a result, the enhanced dataset D_{t+1} becomes more comprehensive and varied, thereby improving the model's generalization capabilities and performance in real-world applications.

Although improving data accuracy and enhancing data diversity have their own emphases in the framework of AIL-ML, they complement each other and jointly improve data quality and model performance. The analysis them reveals several key insights:

- **Complementary Objectives:** Improving data accuracy ensures the reliability of the dataset by minimizing errors and inconsistencies, which is essential for tasks requiring high precision. In contrast, enhancing data diversity broadens the dataset's scope, enabling models to generalize better across varied and unseen scenarios.
- **Different Agent Actions:** Methods aimed at improving accuracy typically involve corrective actions such as correcting mislabeled data, label refinement and removing inconsistencies. On the other hand, diversity enhancement methods focus on generating new samples, manipulating features, creating adversarial examples or transforming existing data through techniques such as data augmentation.
- **Combining both methodologies** leads to a dataset that is not only accurate and consistent but also rich in diversity. High-accuracy data provides a reliable foundation, while diverse data ensures that the model can handle a wide range of inputs, enhancing overall

performance and adaptability.

3.1.4 Data annotation

In the realms of deep learning and machine learning, data annotation is fundamentally crucial due to the dependency of model learning capabilities on extensive, high-quality, annotated datasets. In this section, we introduce AIL-ML methods specifically aimed at enhancing the efficiency and accuracy of data annotation.

Initially, we present an overview of the current practices of agent-based annotation within the AIL framework, outlining how agents contribute to the data labeling process. Then, we explore various methodologies focused on reducing annotation costs and increasing the efficiency of the annotation process. The integration of these strategies ensures that the data annotation is not only cost-effective but also enhances the scalability and applicability of machine learning models in different applications.

3.1.4.1 Mathematical framework for data annotation The agent-assisted data annotation process within the AIL-ML framework can be modeled as an optimization problem, formulated by the following equation:

$$L_t = \gamma(D'_t, A_t, \Sigma_t, \Omega_t) \quad (9)$$

where:

- L_t represents the outputs at time t , which may include labels, confidence levels, and explanations based on the output options Ω_t .
- D'_t denotes the subset of samples selected from the original dataset D_t for annotation.
- A_t is the inputs from the agent.
- Σ_t is a set of annotation strategies, such as fixed-choice, range selection, or open input, and how these strategies are implemented to reduce costs and increase efficiency.
- Ω_t defines the output options, specifying which additional information should be generated.
- γ is the comprehensive annotation function responsible for generating the desired annotated data based on the selected samples S_t , agent inputs A_t , the annotation strategies Σ_t and the output options Ω_t .

This formulation ensures that the data annotation process is not only cost-effective but also scalable and applicable to various machine learning models in different applications. By incorporating intelligent agent strategies, AIL-ML systems can enhance efficiency and reduce costs associated with data annotation.

3.1.4.2 Annotation methods categorized by degree of freedom As illustrated in Fig. 4, the common methods used for annotation can be classified into three types based on the degree of freedom each method provides: fixed-choice, range selection, and open input.

- The fixed-choice method, which offers the least freedom, requires agents to select from at least two predefined options. This is the most commonly used and universally applicable annotation method, offering minimal cognitive burden to agents but lacking flexibility compared to other methods (Wu et al. 2020).
- Range selection, offering more freedom than fixed-choice, includes choosing within discrete or continuous sequences. Discrete sequences consist of clearly distinguishable units, such as text, while continuous sequences, like image pixels or signal durations, lack clear units (Cai et al. 2019; Xu et al. 2022). In these instances, individual data points may not contain complete meaning on their own (e.g., pixels). For some unstructured data, researchers utilize dimension reduction to visualize the data on a graphical interface, allowing agents to select ranges by drawing boundary boxes on spectrograms to manually annotate events (Kath et al. 2023).
- The open input method provides the highest degree of freedom, where agents can input self-defined text. This method enables the generation of rich, diverse annotations, such as dialogues (Hancock et al. 2019) or rules (Zhang et al. 2019). Open input annotations are particularly beneficial for tasks requiring complex data interpretation, allowing agents to offer detailed information that enhances model training and performance.

3.1.4.3 Strategies for reducing annotation costs The reduction of annotation costs within the AIL-ML framework is a prominent research focus. Annotation for certain tasks can be

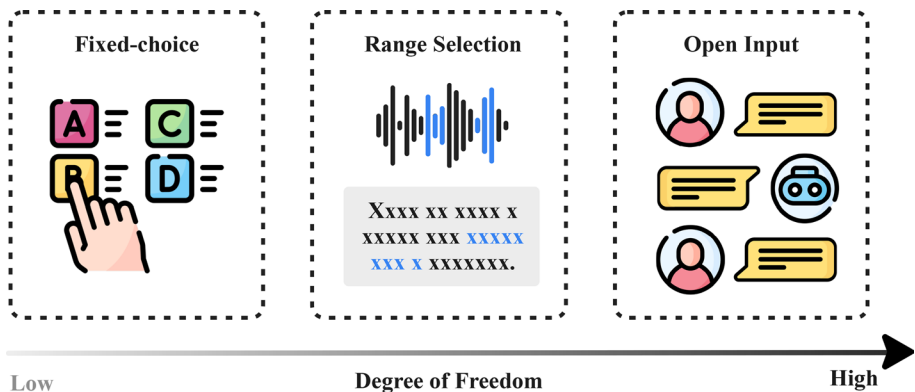


Fig. 4 Classification of annotation methods based on the degree of freedom. The fixed-choice method provides minimal freedom with predefined options for selection. Range selection allows choosing within discrete or continuous sequences, offering more flexibility. The open input method provides the highest degree of freedom, allowing free text input

costly, demanding considerable time, energy, and expertise from annotators. As described in Fig. 5, the AIL-ML framework utilizes three kinds of strategies to reduce annotation costs:

1. **Data:** selecting the most valuable samples for annotation to reduce the number of items needing labels.
2. **Process Optimization:** optimizing the annotation process to enhance efficiency.
3. **Agent:** deploying hierarchical levels of agents to carry out these tasks, thereby diminishing the overall annotation costs.

3.1.4.4 Data: selecting the most valuable samples In the context of enhancing efficiency in machine learning, selecting the most valuable samples for agents to annotate can significantly reduce the number of data points that need to be labeled. One effective approach is the cluster-then-label method, which reduces the volume of data needing annotation by selecting representative samples for labeling, as discussed in the works of Wu et al. (2020) and Hiremath et al. (2022) This approach uses unsupervised clustering to identify key data points that are then annotated, thus diminishing the overall volume of data requiring manual labels. In scenarios where the system can accurately recognize certain activities, it may initiate a one-shot interaction for annotating such behaviors, thereby creating a labeled dataset. Data points with low confidence or high uncertainty in current ML models are deemed highly informative and prioritized for annotation. In parallel, Qian et al. introduced the PARTNER system (Qian et al. 2020), which predicts labels for all unannotated entities using the current model and ranks them based on uncertainty, focusing annotation efforts on the most uncertain instances. Following annotation, similar instances are identified and assigned pseudo-labels based on the annotated data, thereby augmenting the training dataset efficiently. This strategy effectively combines user involvement with model training, minimizing the need for extensive manual annotations. Furthermore, boundary samples present classification challenges due to the inherent features of multiple categories. To address this, Sahu et al. (2023) introduced a method where large models re-annotate boundary samples to ensure accurate classification, thus enhancing both the accuracy and generalization capability of the training models.

All the methods discussed above can be integrated into our proposed Eq. 9. Table 6 illustrates how each approach aligns with specific elements of our framework.

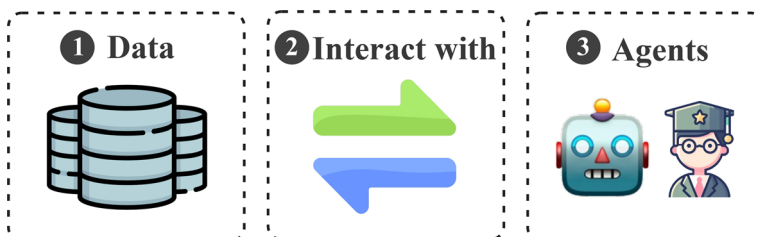


Fig. 5 Three primary strategies for reduce annotation costs within the AIL=ML framework. The strategies include: selecting the most valuable data samples, optimizing data interaction processes and leveraging diverse agents

Table 6 Symbol description for methods selecting the most valuable samples

Method	L_t	γ	D'_t	A_t	Σ_t	Ω_t
Cluster-then-label (Wu et al. 2020; Hiremath et al. 2022)	Labels for selected clusters	A clustering step and a manual labeling step	Cluster centroids as representatives	Human labeling	Fixed-choice	Labels
PART-NER (Qian et al. 2020)	Labels, partial pseudo-labels and confidence scores	Model prediction, user correction and labeling	High-uncertainty instances for user labeling; others pseudo-labeled	Human labeling	Fixed-choice	Label and confidence for pseudo-labeled data
Prompt-Mix (Sahu et al. 2023)	Enhanced labels for classification tasks	Boundary data generation and LLM-based re-labeling	Generated boundary samples via LLM	Data generation and re-labeling	Fixed-choice	Labels with explanations

Table 7 Symbol description for methods optimizing the annotation process

Method	L_t	γ	D'_t	A_t	Σ_t	Ω_t
HIL entity linking (Klie et al. 2020)	Labels for entity linking tasks	Recommendation algorithm and adaptive candidate ranking	Samples selected by recommendation system	Human labeling	Fixed-choice	Labels with recommended scores
LSUN (Yu et al. 2015)	Class labels	Iterative annotation pipeline	Selected via k-means clustering and classifier confidence	Human labeling	Fixed-choice	Labels

3.1.4.5 Process optimization: optimizing the annotation process Optimizing the annotation process to reduce workload and cost while enhancing efficiency is also a key focus in current research. Klie et al. (2020) have effectively utilized recommendation algorithms to propose potential concepts and adaptively prioritize candidates. Their approach has been proven to increase the speed of annotation by 35%, with users expressing a strong preference for this system. Similarly, in the LSUN project led by Yu et al. (2015), the development of an efficient graphical user interface significantly lightens the annotation load and enhances productivity. This interface focuses the annotator on a single image at a time, providing thumbnails of previous and next images to ease understanding and reduce navigation time. Enhancements such as full-screen image display improve the visibility of visual details, leading to more accurate annotations. Moreover, the introduction of simplified keyboard shortcuts, like using the space bar for classification, greatly enhances the efficiency of operations. These interface innovations not only boost the efficiency of data annotation but also substantially lower the costs involved, providing robust support for building extensive image datasets.

Table 7 summarizes how each referenced work maps on Eq. 9.

3.1.4.6 Agent: deploying hierarchical levels of agents In the context of AIL-ML, agents can be categorized into three distinct types: LLMs, general users, and domain experts. As shown in Fig. 6, general users hold general knowledge, which is broad and encompasses a wide array of basic facts across various domains. LLMs, on the other hand, are trained on trillions of data points from extensive databases, encompassing not only general knowledge but also a significant amount of database-specific information. Domain experts stand out by their possession of both general knowledge and highly specialized expert knowledge, specific to their particular field of study. It is important to note that the knowledge capacity of LLMs is limited by the nature of their training data, which is generated by humans and may not cover the depth of expertise that domain experts achieve through specialized education and experience. Thus, the knowledge ceiling of LLMs is typically considered to be below that of domain experts.

Upon training with datasets consisting of trillions of data points, LLMs have developed capabilities that include reasoning, semantic understanding, grounding, and planning. These capabilities suggest that LLMs could serve effectively as proxies for humans in AIL contexts, potentially reducing the substantial costs associated with manual labor (Hsieh et al. 2023a; He et al. 2023b). Researchers like Wang et al. (2021) have explored the use of GPT-3 as a cost-effective tool for data annotation. By generating pseudo-labels through GPT-3, this approach offers a more economical and faster alternative to traditional manual labeling methods. Compared to manual annotation, the use of GPT-3 can reduce costs by 50% to 96%, owing to its ability to generate labels more swiftly and at a lower cost.

Various studies have explored strategies for distributing annotation tasks among different agent types, employing a multi-level agent architecture to lower annotation costs effectively (Wang et al. 2023b; He et al. 2023a). LLMs show considerable efficacy in handling tasks with limited label spaces, significantly reducing manual annotation costs and reaching quality levels akin to human annotations under specific conditions. However, their performance is less robust in scenarios with extensive labeling requirements and complex relationships (Ding et al. 2022). To optimize both cost and quality, integrating LLMs with human annotators proves beneficial, particularly for critical tasks. Shuohang Wang and his team (Wang et al. 2021) have leveraged GPT-3 to generate pseudo-labels and their corresponding con-

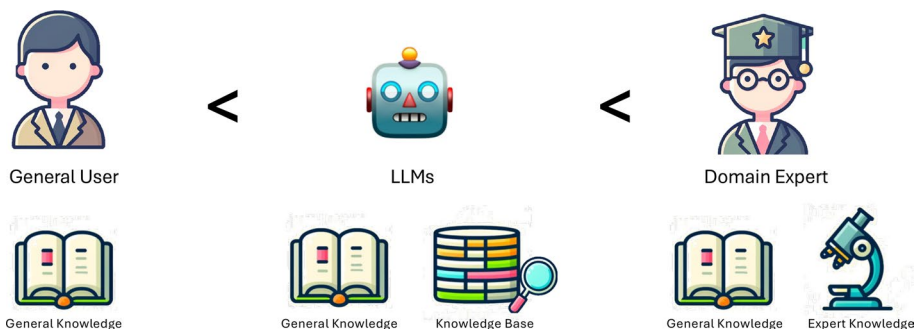


Fig. 6 Three types of agents in the AIL-ML framework: general users, LLMs, and domain experts. General users hold general knowledge across various domains. LLMs are enhanced by training on extensive databases, holding both general and substantial database-specific knowledge. Domain experts, combine general knowledge and expert knowledge, establishing them at the highest level of knowledge within their specific fields

fidence scores, allowing for selective human re-annotation of labels that demonstrate low confidence. This strategy significantly decreases the dependency on human resources without compromising the quality of annotations, which is particularly critical when working within limited budgets on large-scale datasets. While LLMs offer efficiency and cost savings in generating annotations, their performance on complex or domain-specific tasks is typically flawed and may introduce biases relative to human annotations. To address these challenges, Wang et al. (2024) have introduced a collaborative method that utilizes the strengths of LLMs and human annotators to enhance annotation accuracy and reliability at low costs. This approach involves LLMs producing initial labels and explanations, followed by a validation model that evaluates these labels, with human annotators subsequently focusing on the instances scored lowest by the validation model. The LLM-generated explanations provide additional context that aids human annotators in refining and improving the labels, thus cost-effectively ensuring high-quality annotations. Annotating multimodal data, such as text and images from social media, is often challenged by its complexity and noise. To address these issues, Feihong Lu et al. proposed the Miko framework (Lu et al. 2024). This framework leverages the collaboration of multimodal large language models (MLLMs), LLMs, and humans to achieve high-quality annotations. The process starts with MLLMs generating detailed image descriptions to refine and validate textual content, thereby reducing noise and enhancing multimodal data representation. Subsequently, LLMs extract and classify users' potential intentions based on key dimensions such as concepts, actions, and emotions. Finally, human reviewers verify the generated intentions, ensuring their quality and relevance through systematic scoring. By combining the strengths of machine intelligence and human judgment, Miko not only improves annotation efficiency but also enriches models' understanding of user intentions, offering robust support for downstream tasks like recommendation systems and sentiment analysis.

In recent studies, expert agents have been utilized to create rules that significantly decrease the amount of data that needs to be annotated, thereby reducing costs associated with data labeling (Li 2017; Zhang et al. 2019). Zhang et al. (2019) developed a method where human experts use regular expressions to establish initial rules, generating weakly labeled data. While this data might contain noise, it serves as a valuable asset for the early stages of model training, particularly effective when labeled data are scarce. This approach accelerates the annotation process, improving the quality of ML models. However, when there is an abundance of labeled data, deep neural networks excel at uncovering more intricate dependencies in the data. Data derived from regular expressions typically show limited diversity and are strongly rule-oriented, making them suitable primarily in the early stages of model training where labels are deficient. Hemmer et al. (2022) have explored a coordinated approach between classifier systems and expert agents. In their system, classifiers are trained to pinpoint cases that are difficult for human experts to handle, while an assignment system efficiently allocates each case to the most suitable agent, whether human or machine. This approach has been evaluated in several studies, including experiments with "synthetic" experts and a real medical dataset annotated by multiple radiologists. Research indicates that this method outperforms previous approaches and provides more accurate results than either the best human experts or classifiers alone.

All these methods can be incorporated into our proposed Eq. 9. Table 8 provides an overview of how each method corresponds to different elements of our framework.

Table 8 Symbol description for methods deploying hierarchical levels of agents

Method	L_t	γ	D'_t	A_t	Σ_t	Ω_t
GPT3 for Labeling (Wang et al. 2021)	LLM labels and human corrections	Labeling pipeline with GPT-3 inference and human review	Low-confidence subset	GPT-3 predictions (labels, logits) and human corrections	Active labeling	Labels and confidences
Human-LLM Annotation (Wang et al. 2024)	Labels and LLM explanations	LLM labeling and human re-labeling	Low-verification-score data	LLM outputs (labels, explanations) and human corrections	Collaborative labeling strategy	Labels, explanations and verification scores
HITL Entity Extraction (Zhang et al. 2019)	Entity labels	Regex-generated weak labels and annotation on uncertain samples	Uncertainty data	Experts designing regex rules and general user labeling on uncertain samples	Collaborative labeling strategy	Labels with confidence scores
Human-AI Teams (Hemmer et al. 2022)	Collaborative labels	Joint training pipeline for optimal team performance	Dynamic samples	Multiple human experts and AI models	Collaborative labeling strategy	Labels
Miko (Lu et al. 2024)	Image descriptions, key information and user intentions	Multi-stage annotation process	Clean, complete, and multi-modal data (text + images)	MLLM for image descriptions, LLM for key information extraction and intent generation and two-stage human validation	Multi-step annotation strategy	Labels, explanations, and intent classifications

3.1.4.7 Analysis of data annotation in AIL-ML In this section, we introduce three data annotation methods categorized by degree of freedom: fixed-choice, range selection, and open input. Fixed-choice accelerates labeling but may overlook details; range selection enhances granularity at the expense of increased effort; and open input captures deeper insights though it risks inconsistency or noise. For other researchers, an effective choice depends on matching the method's flexibility to the complexity of the data and the goals of the annotation task, suggesting that assessing trade-offs between expressivity and cognitive burden is essential.

Selecting the Most Valuable Sample strategies aimed at minimizing redundant labeling. By identifying the most informative or representative items—for instance, via cluster-then-label methods or uncertainty-based sampling—researchers can focus human resources where they are most needed, reducing overall annotation volume without compromising model performance. This selective approach ensures that each labeled instance contribute to training while preventing time-consuming manual labeling of uninformative samples.

Optimizing the Annotation Process emphasizes that annotation efficiency is not only about data but also the manner in which data is presented and processed. Innova-

tions like dynamic recommendation algorithms, adaptive interfaces, and simplifying agent's workflows can yield substantial gains in speed and consistency. However, implementing these optimizations often requires iterative user testing and interface refinement. This highlights the importance of balancing technical algorithms and agent factors.

Deploying Hierarchical Levels of Agents can utilize the advantage of different agent types. LMs provide rapid, cost-effective pseudo-labeling but may introduce biases or errors if left unchecked, while domain experts can resolve intricate cases at a higher per-annotation cost. General users fill in the middle ground, handling moderately challenging tasks. This layered design encourages thoughtful distribution of effort and expertise but also requires continual calibration—researchers must ensure that each agent level is invoked where it adds value rather than extraneous complexity.

Effective data annotation relies on combining data selection, data sampling, workflow optimization, and appropriate agent collaboration. Each component must work together, such as aligning data selection with interface design or involving experts to handle challenging edge cases. By integrating these elements, researchers can reduce labeling effort and ensure high-quality annotations, ultimately supporting the development of reliable machine learning models.

3.2 Model development and optimization

For machine learning, the development and optimization of models are also important. The methodologies adopted during the model development phase significantly impact the effectiveness and efficiency. Moreover, continual refinement and optimization are essential for the enhancement of model performance. In this section, we investigate AIL-ML techniques that concentrate on model development and optimization. This includes a comprehensive analysis of methods in the areas of *Model Cold Start Issue*, *Model Training*, and *Model Iterative Enhancement*.

3.2.1 Model cold start issues

In machine learning, the model cold start problem refers to the challenge of initializing a model effectively when minimal or no historical data is available. This issue is particularly prevalent for machine learning that is data-driven. The lack of initial training data can significantly lower model performance. In this section, we will explore various AIL-ML methods that address the model cold start problem, aiming to improve the effectiveness when models begin the learning process.

3.2.1.1 Mathematical framework for model cold start issues

Given the initial state

M_0 and minimal or no historical data at time $t = 0$, how to utilize AIL-ML framework to address model cold start issue can be defined as:

$$M_1 = \chi(M_0, \sigma(A_0, P_{\text{gen}}), P_{\text{cold}}) \quad (10)$$

where:

- M_0 is the model's initial state which is usually untrained or only configured.
- $\sigma(A_0, P_{\text{gen}})$ represents the data generation function, influenced by agent's input A_0 and data generation parameters P_{gen} .
- P_{cold} are the parameters specifically designed for rapid adaptation during the cold start phase.
- χ is the cold start learning function, which is responsible for integrating the data generated by the agent with the learning parameters, and quickly updating the state from M_0 to M_1 .

After reviewing the relevant literature within the AIL-ML domain, we have identified two primary approaches to solve the cold start problem: 1) employing agents to generate an initial dataset, and 2) utilizing transfer learning and domain generalization techniques. The first approach involves using agents to generate the data that can serve as a preliminary training set. This leverages the agents' capabilities to simulate realistic data scenarios that help bootstrap the model's learning process. The second approach involves applying knowledge gained from one or more source domains to a target domain where data are scarce. This method not only addresses the scarcity of data in new applications but also enhances the model's ability to generalize across different domains, thus providing a robust foundation for model training even in the absence of extensive domain-specific data.

3.2.1.2 Employing agents to generate an initial dataset During the model cold start phase, a high-quality initial dataset is necessary for model training. Agents can provide or generate initial dataset, thus addressing the cold start problem. Large models can generate the data needed for initial training, enabling smaller task-specific models to begin training without data. Therefore, large models can resolve the cold start issue by supplying or creating initial training data (Ye et al. 2022). Wang et al. (2021) demonstrated how large models can be used to generate annotated data, rapidly assembling an initial dataset to overcome the cold start problem. Especially in low-resource settings, the label data produced by large models provide sufficient information. Large models can shorten the time of acquiring initial annotated data through few-shot learning. Additionally, the cost of using large models for data annotation is significantly lower than that of human annotation, which helps to quickly acquire large-scale high-quality annotated data with limited budgets. These high-quality initial annotations can accelerate the early training process, enabling the model to quickly initiate from a cold start. Besides generating data labels, large models can also produce inferences (Hsieh et al. 2023a). By leveraging inferences and labels generated by large models, smaller models can gain rich contextual information and task knowledge from the start of training. This effectively addresses the cold start problem, allowing small models to quickly achieve better performance with limited training data. Large models significantly reduce the time and financial costs associated with acquiring high-quality datasets, providing a more feasible option for individuals or small organizations to utilize advanced AI capabilities (Ding et al. 2022).

We demonstrate in Table 9 how each referenced study corresponds to the symbols and in Eq. 10.

Table 9 Symbol description for methods employing agents to generate an initial dataset

Method	χ	$\sigma(A_0, P_{\text{gen}})$	P_{cold}
GPT-3 for label- ing (Wang et al. 2021)	Using LM to generate init dataset	Generating dataset by LM in low-resource scenes, providing init training dataset	Few-shot prompting strategies
Distilling Step-by-step (Hsieh et al. 2023a)	Integrating LM's rationales to train the smaller model	Generating rationales from the LM, expanding dataset with step-by-step explanations	Multi-task distillation parameters
GPT-3 for annotation (Ding et al. 2022)	Integrating GPT-3-generated dataset into the training loop to bootstrap the model	Data generation methods (PGDA, PGDG, DADG) producing dataset for cold-start training	Prompt configurations and knowledge-base usage parameters to enhance data quality

3.2.1.3 Utilizing transfer learning and domain generalization techniques The second approach to resolve the model cold start issue involves leveraging transfer learning, where knowledge from one or more well-resourced source domains is applied to a target domain that lacks sufficient data. This method not only addresses the scarcity of data in new applications but also enhances the model's ability to generalize across different domains, thus providing a robust foundation for model training even in the absence of extensive domain-specific data. Similarly, experts can utilize domain generalization to adjust models based on diverse tasks. This method involves learning from one or several related but different domains, allowing the model to generalize well to an unseen domain. Theoretical insights from Jindong Wang and Ben-David support this approach by indicating that invariant feature representations across domains ensure the generalizability and transferability of the model (Wang et al. 2022; Ben-David et al. 2006). Consequently, experts may utilize the pretraining-finetuning strategy within transfer learning to adapt the model to new settings effectively (Zhuang et al. 2020).

3.2.1.4 Analysis of model cold start issues in AIL-ML Employing agents to generate an initial dataset can effectively address the model cold start problem, particularly when working with limited or no labeled data. LLMs can rapidly produce training samples or annotations, effectively training smaller models that would otherwise lack a representative dataset. Although the dataset may contain biases or errors from LLM, this method greatly lowers labeling effort and speeds up model deployment. Moreover, LM have the ability to create large-scale data at low cost, which helps smaller organizations that cannot afford expensive human annotation.

Meanwhile, utilizing transfer learning and domain generalization addresses cold starts from a different angle, emphasizing the transfer of robust, pre-learned feature representations from well-resourced domains to domains with sparse data. By aligning model parameters and feature spaces across different tasks or domains, researchers can equip a newly deployed model with strong priors that enable quick adaptation, even if domain-specific

data are scarce. Transfer learning provides a theoretical and empirical backbone for accelerating cold starts, but its efficacy depends on the compatibility of source and target domains.

LLM-driven dataset creation directly addresses the absence of labeled data by quickly supplying training examples, while transfer learning and domain generalization reuse existing knowledge to guide model parameter initialization. Researchers must balance these approaches according to resource availability, domain similarity, and the potential overhead of data cleaning or adaptation.

3.2.2 Model training

In machine learning, the model training process involves the adjustment of model parameters using optimization algorithms. This procedure involves the selection of the model's architecture, the loss function, and the optimizer to ensure effective learning. The model's architecture influences the model's capacity to capture relevant patterns in the data. Moreover, the loss function ensures that the model's predictions align closely with the actual outcomes. Similarly, the choice of optimizer affects the efficiency and speed with which the model converges to a solution. Agents can leverage their deep understanding of the problem, thereby offering essential insights to enhance model performance.

3.2.2.1 Mathematical framework for model training The model training process, integrating agent's insights into it, can be defined as follows:

$$M_{t+1} = \phi(M_t, D_t, A_t, \Theta_t) \quad (11)$$

where:

- M_t and M_{t+1} represent the model states at time t and $t + 1$, respectively.
- D_t is the training dataset at time t .
- A_t includes inputs from the agent, which may suggest modifications to the model architecture, loss function, or optimization strategy.
- Θ_t encompasses the set of training parameters, which integrates training parameters with the insights provided by the agent.
- ϕ is the model update function that not only applies conventional optimization algorithms but also integrates the agent's guidance to achieve more effective learning and improved model performance.

Generally, AIL-ML framework integration in model training is in two ways: 1) agents use knowledge to optimize model training or 2) agents can be designed as a component of the machine learning network architecture.

3.2.2.2 Leveraging agent knowledge for optimized model learning Agents can leverage their knowledge to guide the design and training of models, ensuring that these models effectively capture the core patterns of the tasks. Agents can influence several aspects of model development, including selecting the appropriate model framework, loss functions, and optimization algorithms, as well as adjusting the training procedures such as the number

of training iterations or epochs (Kath et al. 2023; Tchemeube et al. 2023). Roels et al. (2019) developed an AIL approach to address the explosion of data sets due to 3D technology in electron microscopy (EM). They created an interactive graphical user interface that allows biologists to use the framework in an intuitive and user-friendly fashion. Experts directly interact with the model through this graphical interface to evaluate the performance of various models and choose the most suitable one. Additionally, experts can select regions of interest and set parameters to optimize the model, ensuring the efficiency and reproducibility of the algorithms. A recent study by Metsch et al. (2024) provides a compelling demonstration of integrating interactive graphical interfaces in the context of biomedical data. They introduced the CLARUS platform, an explainable AI tool designed for graph neural networks (GNNs) in the biomedical domain. CLARUS enables experts to manually modify patient-specific protein-protein interaction networks and evaluate the impact of these modifications on GNN predictions through counterfactual analysis. This interactive process not only enhances the understanding of the model's decision-making but also allows experts to retrain the model based on their domain knowledge, improving its interpretability and performance.

For ease of reference, Table 10 compiles the key elements of these approaches in relation to Eq. 11.

3.2.2.3 Designing agents as architectural components in ML Agents can be designed as a component of the machine learning network architecture. These agents continuously provide real-time feedback that adjusts the model's behavior, thereby optimizing the training outcomes. Some researchers incorporated agents directly into the network architecture. Such designs allow the model to learn and adapt during training actively from agents, enhancing model's ability to adjust internal feature representations based on agent inputs Wei et

Table 10 Symbol description for methods leveraging agent knowledge for optimized model learning

Method	ϕ	D_t	A_t	Θ_t
HITL in electron microscopy (Roels et al. 2019)	Executing the de-noising algorithm with specified parameters based on expert input	3D EM image and expert-specified Regions of Interest	Expert interactions via the GUI, including algorithm choice and parameter adjustments	De-noising algorithm parameters such as filter size and iteration count
CLARUS (Metsch et al. 2024)	Integrating expert modifications into the GNN training process, enabling analysis and retraining	Synthetic graph dataset and Patient-specific protein-protein interaction (PPI) networks from gene expression	Expert modifications to the protein-protein interaction networks, such as adding/removing interactions based on domain knowledge	GNN hyper-parameters

al. 2022 Some researchers embedded agents into the network architecture as core elements by the interactive mechanisms and feedback loops (Oh et al. 2019; Weber et al. 2020). Oh et al. (2019) presented a method for interactive video object segmentation where agents annotate video frames (e.g., through scribbles). These annotations are utilized by an interactive network to generate initial masks for foreground objects. The agent's annotations not only serve as input to the interactive network but also continuously influence the network's outputs through multiple rounds of interaction and feedback. Similarly, Wang et al. (2023a) introduced H-Gen, an automatic H-DNN compression framework that incorporates human input as a new hyperparameter for accurate and efficient DNN generation. When the model exhibits low confidence in certain inputs during training, the agent is triggered to generate precise label data for these samples. The H-DNNs generated by H-Gen outperform the original DNNs in terms of accuracy, latency, and energy consumption, significantly improving performance in resource-constrained environments. Large models can also be used as a component of the network (Kwon and Michael 2023; Ye et al. 2023). Kwon and Michael (2023) integrated LLMs into reinforcement learning frameworks as proxy reward functions. They input natural language prompts into the LLM. Then LLM evaluates the behavior of reinforcement learning (RL) agents based on the prompts and produces textual outputs as reward signals. These reward signals are subsequently parsed into numerical reward signals, which are utilized by the RL agents to update their strategies and undertake new training iterations. Although designing effective prompts requires careful consideration, the contextual learning capability of the LLM allows it to capture human-like behavioral priors from minimal examples. This capability facilitates the efficient generation of reward signals that are well-aligned with the user's objectives, thus training RL agents more effectively.

To provide a consolidated view, Table 11 organizes each referenced work according to the parameters in Eq. (11).

3.2.2.4 Analysis of model training in AIL-ML Leveraging Agent Knowledge for Optimized Model Learning demonstrates how agents refine model training. By offering expert-driven adjustments to model architecture, loss functions, and optimization strategies, agents ensure that models are tailored to capture the relevant patterns within complex datasets. This approach allows for more effective training, particularly in specialized domains where automated methods alone may fall short. Integrating agent expertise can lead to more targeted learning processes.

Designing Agents as Architectural Components in ML explores the integration of agents directly into the machine learning pipeline. This approach enables real-time feedback and adaptive learning, allowing agents to continuously influence the model's behavior during training. By embedding agents within the network architecture, models can dynamically adjust their internal representations based on ongoing interactions and feedback.

The effective incorporation of agents into the model training process is many-sided, requiring both the strategic utilization of agent expertise and the architectural embedding of agents within the learning framework. Moreover, this comprehensive approach encourages the development of models that can swiftly adapt to new challenges.

Table 11 Symbol description for methods designing agents as architectural components in ML

Method	ϕ	D_t	A_t	Θ_t
User-guided video segmentation (Oh et al. 2019)	Processing agent feedback to refine and propagate masks with multi-round training	Video frames for object segmentation	Providing annotations(e.g. scribbles) on video frames to guide model refine	Parameters of interactive and propagation networks
H-Gen (Wang et al. 2023a)	Integrating agent into the DNN training process by knowledge distillation	Image	Generating precise labels for low-confidence samples	Parameters for H-DNN compression and optimization
Reward design with LM (Kwon and Michael 2023)	Utilizing the LM's output as reward to update the RL model	User preference text data	Generating reward signals for RL model training	Parameters of RL model and the integration of agent

3.2.3 Model iterative refinement

The development of machine learning models is an iterative process that depends on continuous assessment and refinement. In real-world long-term machine learning systems, data changes over time. This is an important feature to consider, especially in systems that last for months or even years (Chen et al. 2021a). Therefore, when deployed for a long time, models frequent retraining and adjustments to maintain accuracy and relevance. Furthermore, the process of evaluating discrepancies between model predictions and actual outcomes generates crucial feedback, which is fundamental to the iterative enhancement of models. AIL-ML offers robust solutions to address these challenges. In this section, we will explore various AIL-ML methods that specifically focus on model iterative enhancement. AIL-ML methods facilitate the continuous integration of feedback into the learning cycle, enabling models to adapt proactively to changes in data patterns and environmental conditions. This not only helps in fine-tuning the models based on real-time data but also enhances their ability to generalize across varying contexts and over time. By embedding agents that can assess, adjust, and refine model parameters continuously, AIL-ML systems ensure that machine learning models remain effective and responsive to the evolving nature of real-world data, thereby sustaining their performance and relevance in long-term deployments.

3.2.3.1 Mathematical framework for model iterative refinement The iterative enhancement of the machine learning model under the AIL-ML framework can be mathematically represented by the following formula:

$$M_{t+1} = \psi(M_t, D_t, A_t, \Lambda_t) \quad (12)$$

where:

- M_t and M_{t+1} represent the model at time t and $t + 1$.
- D_t is the dataset at time t , which may evolve over time reflecting changes in data patterns.
- A_t denotes the feedback obtained from agent, evaluating the discrepancies between model predictions and actual outcomes.
- Λ_t encompasses a set of parameters guiding the model update process.
- ψ is the model update function that integrates both traditional data-driven updates and agent-based insights to adaptively refine the model in response to changing conditions.

3.2.3.2 Model iterative refinement based on human feedback Incorporating human feedback into model iterations significantly enhances alignment with human intentions, particularly within the field of NLP (Ahn et al. 2023). The use of AIL frameworks is essential in this context, as they enable models to navigate complex linguistic structures, cultural subtleties, and linguistic variations. Without the involvement of human expertise, machine processing tends to be simple and rigid. Retzlaff et al. (2024) further emphasize that reinforcement learning, inherently a Human-in-the-Loop paradigm, greatly benefits from iterative processes integrating human preferences. This perspective aligns with the core principles of AIL frameworks, underscoring the pivotal role of human feedback in refining models and adapting to dynamic environments. For instance, in dialogue tasks, human involvement is crucial to define conversational goals and maintain alignment with expected behaviors, especially in dynamic, user-driven contexts. Recent advancements demonstrate that integrating human preference feedback through reinforcement learning effectively refines models via iterative enhancements (Ziegler et al. 2019; Stiennon et al. 2020; Ouyang et al. 2022; Crochepierre et al. 2022; Fan et al. 2024). This method typically begins with developing a reward function trained on human feedback to capture task-specific priorities. Iterative fine-tuning then adjusts policies to better reflect these priorities. Retzlaff et al. (2024) highlight the critical role of explainability and trust-building mechanisms in such iterative processes, as they enable productive human-agent interaction and ensure models remain adaptable to evolving user needs. For example, Ying Fan et al. proposed using online reinforcement learning to fine-tune text-to-image models. In this setup, the model continually generates new samples and adjusts its generation strategies based on human feedback (Fan et al. 2024). This online and iterative enhancement process heavily relies on the continuous input of human feedback, enabling gradual improvements with each iteration. The model enhances the quality and alignment of the generated images to better reflect human intentions. Direct quality assessments provided by humans train the reward function, thereby increasing both the accuracy and consistency of the model. This feedback assists in handling complex generation tasks, such as creating scenes with multiple objects or specific attri-

butes like color, quantity, and placement, ensuring that the images accurately reflect the text descriptions. Human feedback also significantly extends the model's generalization capability, as it incorporates a broad spectrum of scenarios and descriptions, thus improving the model's adaptability to various applications. Further, human inputs allow for ongoing optimization, helping the model avoid local optima and adapt to new user demands in real-time, reducing training data biases and enhancing the fairness and quality of the outputs. Following a similar methodology, Daniel M. Ziegler et al. have refined language processing tasks with a reward model derived from human preferences, overcoming the constraints imposed by fixed evaluation metrics such as BLEU or ROUGE (Ziegler et al. 2019). This strategy better aligns the model with genuine human linguistic practices and perceptions, thereby enhancing the model's learning signals and exploratory directions, significantly boosting the efficiency of model iterations and the quality of outputs (Crocchepierre et al. 2022).

Models can enhance their performance through active learning and interactive refinement, leveraging human knowledge to identify and correct their gaps, uncertainties, or errors (Wu et al. 2020; Hancock et al. 2019; Qian et al. 2020). Arakawa et al. (2023) proposed PrISM-Tracker, a multimodal procedure tracking framework that employs wearable sensors and state transition data for user-driven error and uncertainty resolution. This framework actively queries users to address uncertainties encountered during tracking, showcasing a method that combines active learning with time series analysis to enhance accuracy. User input allows the model to continually correct errors and optimize performance during practical usage, consistently obtaining high-quality annotated data. By dynamically updating the model, it adapts more effectively to user habits and environmental changes, reducing long-term uncertainties and enhancing the robustness and predictive accuracy of the system. Similarly in the paper (Yao et al. 2019b), the authors improve the performance of semantic parsing by incorporating user feedback, thus enabling iterative model enhancement. In each iteration, the model generates preliminary SQL queries and assesses uncertainties through an error detector, subsequently generating clarifying questions for the user. User feedback is then used to confirm or correct the model's predictions, with the model updating its current state based on this feedback and re-predicting uncertain segments. After multiple iterations, the model progressively adapts to the patterns of user feedback, minimizing unnecessary queries and improving parsing accuracy.

We summarize in Table 12 how the methodologies cited correspond to the variables defined in Eq. (12).

3.2.3.3 Model iterative refinement based on large model feedback Despite the success achieved by model iteration enhancements through human feedback, reliance on human supervision introduces challenges such as the high cost of manual oversight and issues related to quality, reliability, diversity, consistency, and potential biases. In response, researchers have turned to using feedback from LLMs to enhance models for more complex tasks (He et al. 2023a). Canwen Xu et al. have developed a method known as Self-Distillation with Feedback (SDF), which utilizes feedback from LLMs for iterative enhancement and fine-tuning of models (Xu et al. 2023). In this approach, a trained question-answering model generates four distinct responses for each input prompt. These responses are then evaluated by ChatGPT based on usefulness, relevance, accuracy, and detail, with the highest-scoring response selected for further model fine-tuning. Fine-tuning employs newly introduced low-

Table 12 Symbol description for model iterative refinement based on human feedback

Method	ψ	D_t	A_t	Λ_t
Reinforcement learning via human feedback (Ziegler et al. 2019; Stiennon et al. 2020; Ouyang et al. 2022; Crochepierre et al. 2022; Fan et al. 2024)	RL-based iterative update, guided by human preference feedback	Text data; image-to-text data	Providing preference feedback to the output of the model	RL hyper-parameters
PrISM-tracker (Arakawa et al. 2023)	Refining model output based on user feedback	Multi-modal data from wearable sensors (motion + audio) and state transition graph	Correcting the model's predictions at uncertain moments	Parameters for weighting user feedback, query frequency, etc
Model-based interactive semantic parsing (Yao et al. 2019b)	Generating preliminary model outputs, detecting errors, querying the user, and correcting model outputs	Text-to-SQL dataset (WikiSQL, Spider, etc.)	Confirming or correcting uncertain SQL segments	Thresholds for error detection, query frequency, etc

rank adaptation modules that update only the low-rank matrices in the model's linear layers, enabling efficient parameter tuning while avoiding the high computational costs associated with training reward models. This method enhances model performance by achieving fine-grained optimization, capturing subtle feedback differences, and reducing the risk of catastrophic forgetting. It also offers significant advantages in training efficiency and resource utilization. Given that LLMs can partially substitute for human input, some researchers, like Martin Klissarov, have proposed reinforcement learning based on feedback from large models (Klissarov et al. 2023). Klissarov introduced a novel approach called “Motif,” which uses the prior knowledge of LLMs to construct intrinsic reward functions for RL agents, facilitating the learning of preferences. Researchers evaluate model outputs using LLMs, generate preference datasets, and then train reward models to fine-tune the original models. The intrinsic rewards generated by LLM significantly enhance model performance in complex tasks and guide the model to produce behaviors consistent with human intuition.

LLMs have the distinct capability to blend high-level knowledge into specialized models and ensure that model outputs are aligned with human ethical values and intentions (Guo et al. 2024). In this context, Sun et al. (2024) proposed the SELF-ALIGN method, which leverages principle-driven reasoning with the generative strengths of LLMs to enable AI agents to self-align with minimal human intervention. This method employs a set of manu-

ally defined principles to direct the LLM's response generation, greatly diminishing the reliance on human supervisors. The team has developed 16 principles that specify the ideal characteristics of system-generated responses. These principles act as a framework for producing responses that are not only useful but also ethical and dependable. Through a process of fine-tuning, these principles are incorporated into the LLM's parameters, allowing the model to autonomously generate responses that adhere to these guidelines, eliminating the need for direct application of the principles and examples.

These approaches can be systematically integrated into the proposed framework. Table 13 details how each method relates to the terms of Eq. (12).

3.2.3.4 Analysis of model iterative refinement in AIL-ML In Model Iterative Refinement, the kernel challenge lies in keeping a model's performance and adaptability over extended deployments. Equation (12) highlights how a model transitions from one state

M_t to another M_{t+1} by assimilating new data D_t and agent feedback A_t under specific update parameters Λ_t . By integrating feedback directly into the update loop, models evolve in response to both shifts in data distributions and performance gaps identified during real-world usage.

Model Iterative Refinement Based on Human Feedback refines model behavior and resolve ambiguities, harnessing a key strength of humans to make context-sensitive judgments. Although manual correcting can be costly or easy to variability, it also delivers a high degree of interpretability and domain expertise. This feedback loop is particularly potent in tasks where purely algorithmic metrics (e.g., BLEU scores in language tasks or standard accuracy measures) fail to capture domain-specific or user-centric nuances. Iterative refine-

Table 13 Symbol description for model iterative refinement based on large model feedback

Method	ψ	D_t	A_t	Λ_t
Baize (Xu et al. 2023)	Iterative self-distillation process selecting highest-scoring response, then fine-tuning via LoRA	Generated candidate responses	Ranking candidates to use the best response for model fine-tuning	LoRA adaptation parameters
Motif (Klissarov et al. 2023)	Updating RL model using a combined reward (intrinsic from LLM + extrinsic from environment)	Environment data	Providing preference feedback to the output of the model	RL hyperparameters
SELF-ALIGN (Sun et al. 2024)	Principle-driven self-alignment procedure incorporating the 16 principles	Expert-defined principles and seed prompts	Principle-driven feedback by LM (replacing extensive human supervision)	Parameters or strategies for incorporating these principles into the model

ment not only improves immediate performance but also reveals new data facets or user needs over time, leading to gradually richer and more accurate models.

Model Iterative Refinement Based on Large Model Feedback-where LLMs supply iterative feedback signals-extends scalability by partially or entirely substituting human. This approach retains a continuous learning loop but relies on LLMs as proxies for human judgment, offering faster turnaround and reduced manual overhead. Although such model-generated feedback can accelerate refinements, it also introduces potential biases or knowledge gaps originating from the LLM itself. Therefore, researchers need to balance the convenience and scalability of automated feedback against the risk of recursive error propagation, ensuring that checks and balances remain in place to maintain reliability and ethical alignment.

4 Overview of application domains

4.1 Applications in general knowledge domain

In the domain of general knowledge within AIL-ML, various innovative applications have been developed to enhance computational experiences, making them more enriching, helpful, and adaptable to specific deployment environments.

In the context of intelligent home environments, the incorporation of AIL-ML is gradually reshaping the interaction dynamics between users and automated systems, enabling these systems to better understand and adapt to user behaviors through iterative feedback loops. As shown in Fig. 7, this adaptation process leverages real-time data and agent inputs to continuously refine and optimize the performance of smart home devices. Bootstrapping methods for Human Activity Recognition in smart homes initiate with passive observation, gradually building activity models through minimal supervision, thereby recognizing frequent activities with significant precision (Hiremath et al. 2022). Additionally, applications such as the “Listen Learner” (Wu et al. 2020) leverage user interactions for activity recognition, minimizing user effort while enhancing system adaptability to specific environments. The PrISM-Tracker framework (Arakawa et al. 2023) employs wearable sensors and graph-based state transitions, coupled with user inputs, to refine procedure tracking, addressing uncertainties in real-time. The AI-to-Human Actuation (Cho et al. 2023) approach actively makes user modify environmental conditions to improve the robustness of visual AI sensors, aligning system responses with human activities more accurately. Additionally, Jiayuan Gao et al. proposed LLMIE-UHAR (Gao et al. 2024), a method that leverages LLMs and iterative evolution to achieve unsupervised HAR. In an IoT environment, multiple smart devices generate vast streams of data. These data not only consist of raw information but also contain rich contextual and semantic information, such as the background, time, location, and interactions between devices. LLMIE-UHAR uses prompt engineering to transform sensor data into textual descriptions rich in contextual information, making it comprehensible to LLMs. Then, the LLM is used for analysis and inference, enabling precise annotation of the sensor data. These applications highlight the integration of agent insights with machine learning algorithms to enhance smart home ecosystems.

In the intelligent dialogue systems domain, the focus is on refining conversational agents to offer more human-like interactions through advanced natural language processing and

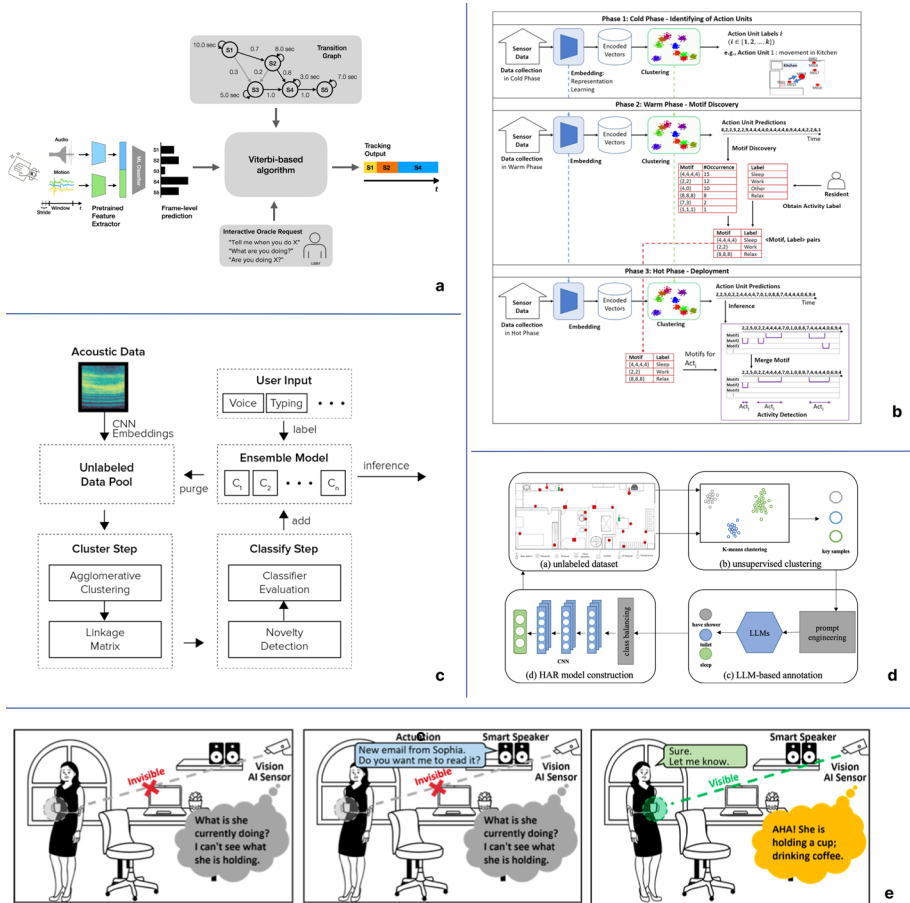


Fig. 7 Frameworks of AIL-ML application in smart home environments. **a** Arakawa et al. (2023): Overview of PrISM-Tracker architecture which utilizes a transition graph and user-provided simulated oracles to enhance the robustness and accuracy of HAR systems in procedural tracking. **b** Hiremath et al. (2022): Overview of our bootstrapping method. Phase 1 involves representation learning and clustering to identify action units. Phase 2 focuses on discovering higher-level features, and Phase 3 involves deploying the activity detection method within the smart home environment. **c** Wu et al. (2020): Listen Learner architecture, a self-supervised algorithm that detects salient acoustic events and generates classifiers for activity recognition, effectively minimizing user involvement. **d** Gao et al. (2024): Overview of LLMIE-UHAR that leverages large language models and iterative evolution to realize unsupervised HAR. **e** Cho et al. (2023): An example of how AI-to-human actuation complements an AI sensor under perception difficulty. Note: The figures provide a rough conceptual understanding, and the text is not intended to be read in full detail. The figures are borrowed from the following papers: (a) Arakawa et al. (2023), (b) Hiremath et al. (2022), (c) Wu et al. (2020), (d) Gao et al. (2024), (e) Cho et al. (2023)

context management. The use of AIL-ML is key, enabling these systems to learn from user interactions and improve over time (Hancock et al. 2019). AIL-ML can not only efficiently utilize real-time conversational data for training but also dynamically adjust based on user feedback and satisfaction, leading to an improvement in both the overall model performance and user experience.

4.2 Applications in specialized knowledge domain

4.2.1 Medicine and healthcare

The significance of AIL-ML is increasingly recognized in specialized fields such as medicine or healthcare (Budd et al. 2021). Despite the promising capabilities of large language models in medicine, the enhancement of Clinical Decision Support systems and complex clinical areas requires the incorporation of human insights through AIL methodologies. As shown in Fig. 8a, in (Liu et al. 2023), clinical decision support logic summaries were provided to ChatGPT, an AI-based question-and-answer tool developed from large language models, to generate recommendations. These AI-generated recommendations were reviewed by multiple clinicians and found to be high in understanding and relevance. However, they scored lower in terms of utility, acceptance, bias, inversion, and redundancy. In applications like Medical Decision-Making, particularly with image retrieval systems, machine learning algorithms fall short of capturing the nuanced understanding of similarity as perceived by experts. This gap led to the development of a tool named SMILY (Cai et al. 2019). It can be observed from Fig. 8 that the tool allows clinicians to directly influence search algorithms, translating certain parameters into refined medical concepts for enhanced diagnostic effectiveness and algorithmic trust without sacrificing accuracy. This approach not only improves the diagnostic process but also enables users to adopt innovative strategies for a deeper understanding and rectification of both machine learning and human errors.

The application of AIL-ML extends to dealing with computationally intensive tasks such as subspace clustering, protein folding, and the K-anonymization of health data. In these scenarios, the exponential search space can be significantly reduced with the aid of human expertise through the heuristic selection of samples. In the context of health informatics, particularly when tackling limited datasets or rare occurrences, the limitations of automated machine learning due to insufficient training samples become evident. A study advocates for the adoption of interactive machine learning, described as algorithms that enhance their learning outcomes through interactions with agents, including humans (Holzinger 2016). This approach allows AIL-ML to utilize human knowledge and experience, optimizing the learning trajectories of machine learning models and thereby elevating their accuracy and efficiency.

The deployment of AIL-ML in disease diagnosis, such as Coronary Artery Disease, leverages expert insights integrated into the classification process, enhancing diagnostic accuracy (Samaras et al. 2023). This approach not only augments the interpretability and transparency of the models but also bolsters their reliability. AIL-ML applications effectively address data bottlenecks in the treatment of various diseases, including pancreatic cancer. According to Fig. 8c, by incorporating nuanced human judgments and expertise, these systems enhance the efficiency and outcomes of medical interventions, as demonstrated in recent studies (Mosqueira-Rey et al. 2023b).

4.2.2 Finance

AIL-ML is a paradigm that integrates agents into the machine learning loop, making it particularly suitable for fields that require strict supervision, such as finance. The application of

Decision Support Editors

BestPractice Advisory: Immunocompromised and Live Virus Immunization

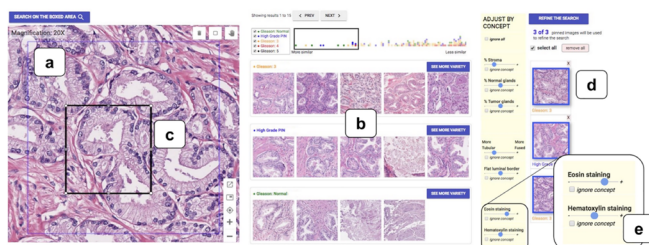
Criteria

Inclusion:
 (Patient has immunosuppressed diagnosis, HIV, primary immunodeficiency
 OR
 CD4 count less than 200
 OR
 Patient has Corticosteroids, Methotrexate, Cyclosporine, Tacrolimus, Sirolimus, Mycophenolate, Cyclophosphamide, Anti-thymocyte globulin, Azathioprine, Mycophenolate Mofetil on the medication list
 OR
 Patient has live vaccine allergy)
 AND
 Patient has a live vaccine order

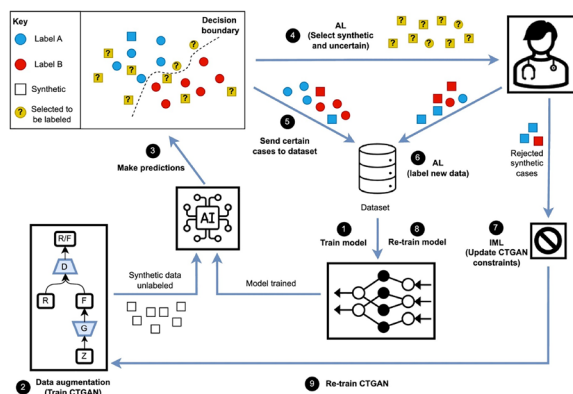
AI-generated Suggestions

- Exclude: Patients who have recently undergone bone marrow transplant or solid organ transplant
- Exclude: Currently receiving immunosuppressive therapy or who have recently completed such therapy
- Exclude: Patient is pregnant or breastfeeding
- Exclude: Patient has a history of cancer
- Exclude: Patient is currently receiving radiation or chemotherapy treatment.
- Some examples of additional medications or treatments that may have immunosuppressive effects include certain cancer treatments, such as chemotherapy or radiation therapy, as well as certain medications used to treat autoimmune disorders, such as rituximab or infliximab.
- Biologic agents, such as adalimumab, etanercept, and golimumab, which are used to treat autoimmune disorders.
- Medications used to treat transplant rejection, such as basiliximab, daclizumab, and muromonab-CD3.

a



b



c

Fig. 8 Frameworks of AIL-ML application for specialized knowledge domain. **a** Liu et al. (2023) A prototype of potential implementation in EHR system-AI decision support editors. **b** Cai et al. (2019) Key components of SMILY. **c** Mosqueira-Rey et al. (2023b) The workflow of the AITL system which is designed for expert to resolve the data bottleneck problem for the treatment of pancreatic cancer. Note: The figures provide a rough conceptual understanding, and the text is not intended to be read in full detail. The figures are borrowed from the following papers: (a) Liu et al. (2023), (b) Cai et al. (2019), (c) Mosqueira-Rey et al. (2023b)

AIL-ML in finance is growing in importance since it enhances the safety and interpretability of ML models.

Despite the growing use of ML in finance, the "black box" problem remains a major barrier. Specifically, the "black box" is the risk that ML models' outputs and operations are unknown to and uncontrolled by human. Buckley et al. (2021) suggest that the most effective way to address this black box issue is by bringing humans into the ML loop, strengthening internal governance where external financial supervision may not be sufficient. They propose three key AIL tools: (1) AI due diligence, which involves a thorough human evaluation of potential risks before the development and deployment of AI; (2) AI explainability, which ensures transparency in the model's decision-making process, making it understandable to humans; and (3) AI review committees, where human oversight is responsible for ethical decision-making, ensuring accountability for AI actions.

4.2.3 Law

Legal services is important in protecting individual rights and maintaining social fairness (Auerbach 1977; Tushnet 2009; Cui et al. 2024). However, these services face many challenges, such as high costs, limited resources, complex legal terminology, frequent legislative changes and so on. Despite the impressive performance of large models in various domains, they encounter the "hallucination problem" in the legal field, where generated content may be inaccurate or outdated, which will cause legal risks.

A notable example of applying AIL-ML in the legal domain is Chatlaw (Cui et al. 2024), a multi-agent collaborative framework designed for intelligent legal consultation. As shown in Fig. 9, agents are involved at multiple stages of the system. During the data preprocessing stage, human refine and annotate the cleansed dataset, enabling the construction of high-quality knowledge graphs and question-answer pairs. Furthermore, agents are integrated into the core workflow of Chatlaw. Multiple agents, such as legal assistants and senior lawyers, interact with users, dynamically updating knowledge graph nodes, generating legal advice, and validating the relevance and legality of retrieved legal cases and clauses. This process ensures the generated legal advice is accuracy and authority. Chatlaw not only addresses the inherent interpretability challenges of legal tasks but also offers valuable insights for developing reliable and context-aware legal support systems.

5 Discussion about limitation and future work

5.1 Limitation

AIL-ML, as a process-oriented paradigm, spans multiple research domains but lacks well-defined research objectives and standardized methodologies characteristic of fields like transfer learning or domain generalization. While it is widely applied in areas such as data acquisition, model training, cold-start problems, and data annotation, AIL-ML incorporates diverse techniques, including human feedback, active learning, and LLMs. Due to the broad range of applications and the variety of techniques involved, AIL-ML methods are challenging to describe in a precise and detailed manner. Instead, they can be summarized by focusing on optimization strategies in data acquisition, processing, and model development,

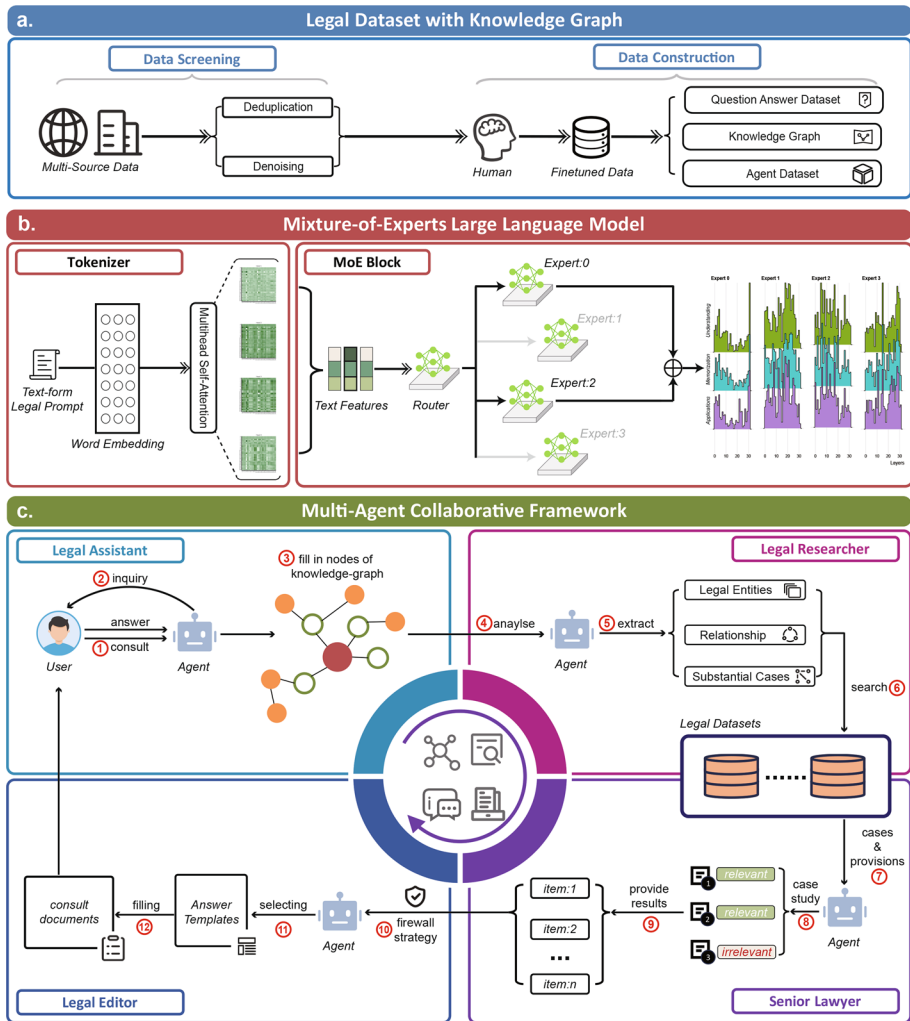


Fig. 9 Framework of Chatlaw (Cui et al. 2024). Note: The figure is borrowed from Cui et al. (2024)

such as reducing annotation costs, improving data initialization, or enhancing model training through agents. This high-level abstraction provides flexibility and broad applicability but also introduces the limitation of lacking standardization in research methods. Consequently, much of the ongoing work focuses on integrating existing techniques within the AIL-ML framework, without establishing a unified technical process or evaluation criteria. Therefore, AIL-ML functions more as a guiding philosophy than as a specific technical toolkit. Future research should prioritize the development of domain-specific methodologies and standardized frameworks to enhance the consistency and practicality of AIL-ML applications.

A potential limitation of the AIL-ML framework lies in its reliance on the performance of LLMs. While LLMs provide advanced capabilities in reasoning and language under-

standing, their outputs in specialized domains may be inaccurate or misleading, potentially lowering the overall system performance. Additionally, LLMs are often considered "black boxes," lacking transparency and interpretability. This limitation can hinder the adoption of AIL-ML in fields requiring high levels of trust and explainability, such as medical diagnostics. AIL-ML relies on a combination of human input and agent feedback, making it more dependent on the quality and reliability of the agents used within the system. This distinction highlights the need for further research into improving the transparency and accuracy of LLMs when integrated into the AIL-ML framework.

5.2 Future directions

Building Trust in AIL-ML Systems with Counterfactual Explanations One critical aspect of AIL-ML systems is establishing trust between agents and the models. Trust can be achieved through counterfactual explanations, which help users understand how and why a model's output changes under hypothetical input conditions. This aligns with human reasoning processes, enabling non-experts to familiarize themselves with the decision boundaries of ML models. Del Ser et al. (2024) conduct a generative framework that employs counterfactual explanations to balance plausibility, change intensity, and adversarial power through multi-objective optimization. This approach not only enhances interpretability but also uncovers concept-based biases and misrepresented features within the model, making it a valuable tool for fostering trust. This facilitates better understanding of model limitations, encourages informed decision-making, and ensures that AIL-ML systems are perceived as reliable and transparent. Future research could explore combining counterfactual explanations with other interpretability techniques to create a more comprehensive trust-building framework, further advancing AIL-ML's applicability in sensitive and high-stakes environments.

Reducing Bias in AIL-ML Framework. In AIL-ML, how to reduce bias is an important and valuable research direction. Currently, both LLMs and human experts are inherently biased. For instance, LLMs may exhibit selection bias by being overly sensitive to changes in input options' order or position (Zheng et al. 2023) and the training data for LLMs often contains embedded social, cultural, or historical biases that can be amplified in outputs. Additionally, human experts involved in the AIL process may also have cognitive biases, which can result in unbalanced or biased knowledge being introduced into the ML. Therefore, future research must focus on developing methods to ensure unbiased knowledge injection within the AIL-ML framework. This involves not only mitigating bias at the source (from both agents like LLMs and human contributors) but also establishing mechanisms to detect and correct biases during data annotation, model training, and feedback loops. Approaches could include designing fairer selection mechanisms, implementing diverse annotation strategies, and employing de-biasing techniques to counteract both selection and data biases. Moreover, algorithms capable of identifying and rectifying potential biases in human and LLM-based annotations will further enhance the fairness and reliability of AIL-ML systems. By systematically addressing bias, AIL-ML can improve its applicability to high-stakes domains such as healthcare and legal decision-making, where trust and explainability are important, while also promoting fairness and generalizability across diverse applications. This research direction is crucial not only for enhancing the technical performance of AIL-ML systems but also for advancing the social responsibility of AI technologies.

Enabling Agents to Handle High-Dimensional and Unstructured Data in AIL-ML. A significant research direction for AIL-ML is finding effective methods to enable agents to process high-dimensional and unstructured data. Currently, most AIL-ML systems are designed to handle low-dimensional data or natural language, with limited exploration of unstructured data types such as images, sensor signals, and acoustic data. In practical applications like smart homes and healthcare, the lack of context-awareness for these unstructured data types limits the potential of intelligent systems. Addressing this challenge by developing techniques that allow agents to handle complex, high-dimensional data can unlock new possibilities for AIL-ML, enabling more robust contextual understanding and improved performance in diverse environments. This direction holds great promise for enhancing the adaptability and intelligence of systems in fields like IoT and smart healthcare, where dealing with unstructured data is crucial.

Enhancing Collaboration Between Large Models and Humans in AIL-ML. A promising future direction for AIL-ML research is focusing on improving the collaboration between LLMs and humans. As AIL systems continue to evolve, the autonomy and collaboration mechanisms in multi-agent systems become critical. Future research can explore how to establish effective collaboration frameworks among multiple agents, including domain-specific LLMs and human experts. These agents can collaborate and share knowledge, enabling more effective handling of complex tasks. By fostering better cooperation between large models and human experts, AIL-ML systems can significantly enhance their performance and adaptability across diverse applications. This direction opens up new possibilities for optimizing human-AI interaction, making AIL systems more versatile and capable in various real-world scenarios.

6 Conclusion

Human-in-the-Loop Machine Learning (HIL-ML) has become an important research topic in machine learning research. The development and popularization of large models introduce both challenges and opportunities to this field. In this paper, we present a comprehensive survey of AIL-ML (which is also called HIL-ML), providing an in-depth review of its methodologies across key stages of machine learning, including a structured review of data collection, data initialization, data quality enhancement, data annotation, model cold start, model training, and iterative model refinement. Furthermore, we summarize AIL-ML applications in diverse fields. Finally, we analyze the current challenges and highlight future directions. We hope that this survey will serve as a valuable resource for researchers, offering insights that encourage further exploration and development in AIL-ML.

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Author contributions Jiayuan Gao conducted the literature review, developed the framework, wrote the manuscript, drafted the figures, prepared Figures 6,7,8 and coordinated the work among the authors. Yingwei Zhang and Yiqiang Chen contributed to the framework development. Yihan Dong conducted the literature review and prepared Figures 1 and 2. Yuanzhe Chen contributed to writing the manuscript and prepared Figure 3. Shuchao Song prepared Figures 4 and 5. Boshi Tang and Yang Gu provided suggestions for manuscript revisions. All authors reviewed the manuscript.

Data availability No datasets were generated or analysed during the current study.

Declarations

Conflict of interest The authors declare no Conflict of interest.

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Authors and Affiliations

Jiayuan Gao^{1,2} · Yingwei Zhang^{1,2} · Yiqiang Chen^{1,2} · Yihan Dong³ · Yuanzhe Chen^{1,2} · Shuchao Song^{1,2} · Boshi Tang⁴ · Yang Gu^{1,2}

✉ Yiqiang Chen
yqchen@ict.ac.cn

Jiayuan Gao
gaojiayuan20z@ict.ac.cn

Yingwei Zhang
zhangyingwei@ict.ac.cn

Yihan Dong
3120215895@bit.edu.cn

Yuanzhe Chen
chenyuanzhe21s@ict.ac.cn

Shuchao Song

songshuchao22b@ict.ac.cn

Boshi Tang
tbs22@mails.tsinghua.edu.cn

Yang Gu
guyang@ict.ac.cn

- ¹ Beijing Key Laboratory of Mobile Computing and Pervasive Device, Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China
- ² University of Chinese Academy of Sciences, Beijing, China
- ³ Beijing Institute of Technology, Beijing, China
- ⁴ Tsinghua University, Beijing, China